SYSTEM REQUIREMENTS SPECIFICATION
STUDENT MANAGEMENT SYSTEM

Version 1.0

Document Change Log

Ngày	Phiên bản	Mô tả	Tác giả
20/10/26	1.0	Initial document	
30/10/26	1.1	Update Use-case Login	
31/10/26	1.2	Update requirement	

I. Functional Requirement

The university needs a student management system with the following basic features:

1. Account Management

- Students, Teachers, and the Administrator log in and out of the system. Students log in with their student ID and password. Teachers log in with teacher ID or email and password.
- Students can view their information and update only their contact details (email, phone number, address). Teachers can update their phone number and email. Other information can only be changed by the Administrator.
- If a user forgets the password, they can request a reset via email or ask the Administrator.
- The Administrator can create, edit, lock, and unlock Student and Teacher accounts.
- Access is restricted by role: each user can only use the functions and see the data they are permitted to.

2. Student Information Management

- The Administrator adds student records: student ID, full name, date of birth, gender, email, phone number, address, place of birth, academic cohort, major, and student class.
- The Administrator can edit student information and academic status (Enrolled, Withdrawn, Temporarily Suspended, Graduated).
- The Administrator can view all students. Teachers can view students in the course sections assigned to them.
- The system shows clear messages when an operation fails.

3. Student Search And Filtering

- The Administrator and Teachers can search students by student ID, full name (partial or complete), student class, or major. Teachers only see students in their assigned course sections.
- The Administrator and Teachers can filter by student class, academic cohort, major, and academic status.
- Results show student ID, full name, student class, major, and academic status. If nothing matches, the system shows a notification.
- The Administrator and Teachers can also open a student's detailed record from the results.

4. Course and course section management

- The Administrator manages courses: course code, course name, credit hours, number of class periods. Course codes must be unique.
- The Administrator creates course sections for each semester: course, class schedule, maximum capacity, and assigned Teacher.
- The Administrator can view the student list of any course section. Teachers can view the lists of their assigned sections.
- Students, Teachers, and the Administrator can view the course sections offered each semester

5. Course registration management

- Students view the course sections open for registration, including course name, code, credit hours, Teacher, schedule, and remaining capacity.
- Students register for a course section during the registration period. The system checks the registration period, remaining capacity, duplicate registration, and prerequisites. Invalid requests are rejected with a reason.

- Students can view their registered sections and cancel a registration before the cancellation deadline.
- The system updates remaining capacity after each registration or cancellation and ensures the number of students never exceeds the maximum capacity

6. Grade and academic result management

- Assigned Teachers enter and edit grades for Students in their course sections while the gradebook is not locked.
- Grade components: attendance, midterm, and final examination, each with a weight (total 100%). The system automatically calculates the final course grade.
- The system converts the final grade into a letter grade (A, B, C, D, F) and a four-point scale, and calculates the semester GPA (average of four-point grades weighted by credit hours).
- The system checks that grades are within the valid range before saving.
- The Teacher can lock the gradebook after grading is finished. Only the Administrator can unlock it.
- Students can view their own component grades, final grades, and semester GPA. Other users cannot view or change them without permission.

II. Non-Functional Requirement

1. Usability

- The system shall provide a simple and user-friendly interface.
- The system shall provide clear instructions for users to perform each function.
- The system shall display clear error messages when users enter invalid information.
- The system shall allow users to navigate between functions easily.
- The system shall provide an appropriate interface for each user role: Administrator, Teacher, and Student.

2. Reliability

- The system shall store student information, course information, and academic results accurately.
- The system shall validate input data before saving it to the database.
- The system shall prevent duplicate student IDs and course codes.
- The system shall display an error message when an operation fails.
- The system shall not modify existing data if an update operation fails.

3. Performance

- The system shall respond to common operations, such as login, student search, and viewing academic results, within 3 seconds under normal operating conditions.
- The system shall support 500 users accessing the system simultaneously.

4. Security

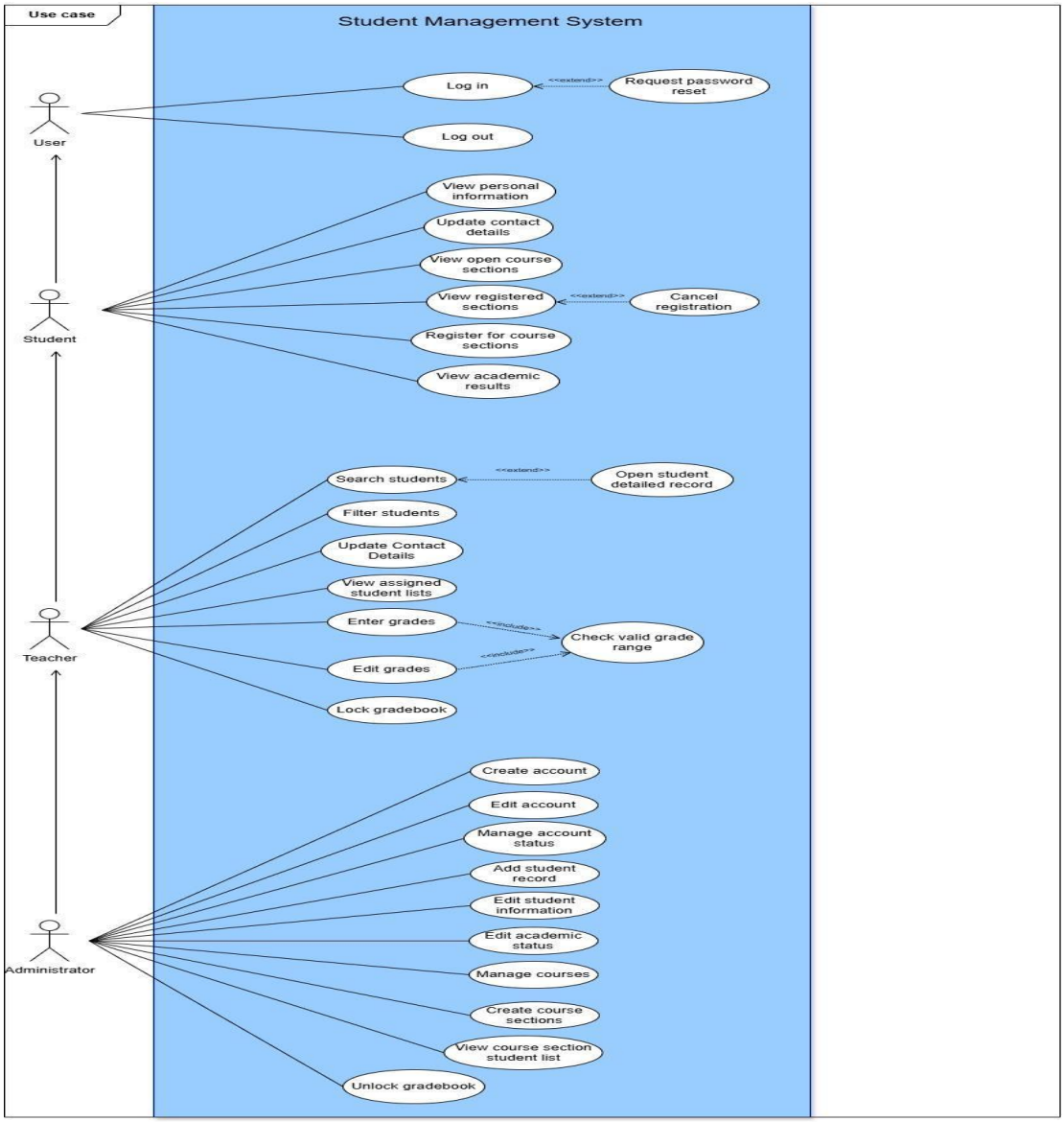
- The system shall assign access permissions according to user roles.
- The Administrator shall be allowed to manage student information, course information, and user accounts.
- The Teacher shall be allowed to manage grades only within the permissions granted to them.
- The Student shall only be allowed to view their own academic results.

5. Development

- The system shall be developed using the Client–Server architecture, in which clients send requests to the server and the server processes those requests.
- The client shall communicate with the server to perform functions such as logging in, searching for students, managing courses, and viewing academic results.
- The server shall process business logic and communicate with the database to retrieve or update data..
- The server shall validate input data and verify user permissions before performing operations involving system data.

III. Use-Case Diagram

1. Overview



1.1. List of Actors

No.	Actor	Description
1	User	Generalized actor representing any person who uses the system (Student, Teacher or Administrator); logs in, logs out and can request a password reset
2	Student	A student who has an account and is logged in; views personal information, registers for course sections and views academic results
3	Teacher	A teacher who views assigned student lists, enters and edits grades, and locks the gradebook
4	Administrator	The system administrator who manages accounts, student records, courses and course sections, and unlocks gradebooks

1.2. List of Use-cases

No.	Use-case	Description
1	Log in	Log in to the system
2	Log out	Log out of the system
3	Request password reset	Request a password reset
4	View personal information	View personal information
5	Update contact details	Update contact details

6	View open course sections	View open course sections
7	Register for course sections	Register for course sections
8	View registered sections	View registered course sections
9	Cancel registration	Cancel a course section registration
10	View academic results	View academic results
11	View assigned student lists	View the lists of assigned students
12	Filter students	Filter students
13	Open student detailed record	Open a student's detailed record
14	Enter grades	Enter grades
15	Edit grades	Edit grades
16	Check valid grade range	Check that grades are within the valid range
17	Lock gradebook	Lock the gradebook
18	Unlock gradebook	Unlock the gradebook
19	Create account	Create an account
20	Edit account	Edit an account
21	Manage account status	Manage account status
22	Search students	Search for students
23	Add student record	Add a student record
24	Edit student information	Edit student information
25	Edit academic status	Edit academic status

26	Manage courses	Manage courses
27	Create course sections	Create course sections
28	View course section student list	View the student list of a course section

2. Use-case Log in

2.1. Summary

This use-case describes how a user (Student, Teacher or Administrator) logs in to the Student Management System.

2.2. Flow of events

Main flow

The use-case starts when a user wants to log in to the system.

1. The system asks the user to enter a username and password.
2. The user enters the username and password and requests to log in.
3. The system checks the entered information. If it is correct, the system lets the user log in and displays the functions that match the user's role (Student, Teacher or Administrator).

Alternative flows

- Wrong username or password: the system displays an error message. The user can go back to the start of the main flow or cancel the login, at which point the use-case ends.
- Account locked or inactive: the system reports that the account cannot be used and the use-case ends.

2.3. Special requirements

Passwords must be stored in encrypted form.

2.4. System state before the use-case starts

The user is not logged in.

2.5. System state after the use-case ends

If the use-case succeeds, the user is logged in. Otherwise, the system state does not change.

2.6. Extension points

Use-case Request password reset: if the user has forgotten the password.

3. Use-case Log out

3.1. Summary

This use-case allows a logged-in user to leave the system.

3.2. Flow of events

Main flow

The use-case starts when the user selects the log out function.

1. The system ends the user's session.
2. The system redirects the user to the login page.

Alternative flows

None.

3.3. Special requirements

None.

3.4. System state before the use-case starts

The user must be logged in.

3.5. System state after the use-case ends

The user is not logged in.

3.6. Extension points

None.

4. Use-case Request password reset

4.1. Summary

This use-case allows a user (Student, Teacher or Administrator) to request a password reset when the password has been forgotten. It extends the use-case Log in.

4.2. Flow of events

Main flow

The use-case starts when the user selects the password reset function while logging in.

1. The system asks the user to enter the registered email or username (student ID or teacher ID).
2. The user enters the information and submits the request.
3. The system checks whether the information exists in the system. If it does, the system sends instructions or a new password to the user's email.
4. The system reports the result of the request.

Alternative flows

The information does not exist in the system: the system displays an error message. The user can re-enter the information or exit the function.

4.3. Special requirements

The new password must be randomly generated and stored in encrypted form.

4.4. System state before the use-case starts

The user is not logged in.

4.5. System state after the use-case ends

If the use-case succeeds, the user's password is reset. Otherwise, the system state does not change.

4.6. Extension points

None.

5. Use-case View personal information

5.1. Summary

This use-case allows a student to view their own personal information.

5.2. Flow of events

Main flow

The use-case starts when the student selects the function to view personal information.

1. The system displays the student's personal information: student ID, full name, date of birth, gender, address, phone number and email.

Alternative flows

None.

5.3. Special requirements

A student can only view their own information.

5.4. System state before the use-case starts

The student must be logged in.

5.5. System state after the use-case ends

None.

5.6. Extension points

None.

6. Use-case Update contact details

6.1. Summary

This use-case allows a student or a teacher to update their contact details (student: address, phone number, email; teacher: phone number, email).

6.2. Flow of events

Main flow

The use-case starts when the user selects the function to update contact details.

1. The system displays the user's current contact details.
2. The user edits the contact details as desired.
3. The user requests the update.
4. The system validates the information and updates it in the system.
5. The system reports the result of the update.

Alternative flows

- The user cancels: the system does not update the information and the use-case ends.
- Invalid information or missing required fields: the system displays an error message. The user can correct the information or cancel the update.

6.3. Special requirements

A student can update address, phone number and email; a teacher can only update phone number and email. Neither can change academic information.

6.4. System state before the use-case starts

The user (Student or Teacher) must be logged in.

6.5. System state after the use-case ends

If the use-case succeeds, the contact details are updated. Otherwise, the system state does not change.

6.6. Extension points

None.

7. Use-case View open course sections

7.1. Summary

This use-case allows a student to view the list of course sections that are open for registration.

7.2. Flow of events

Main flow

The use-case starts when the student selects the function to view open course sections.

1. The system displays the list of open course sections: section code, course name, teacher, schedule, maximum capacity and remaining seats.

Alternative flows

No course section is open: the system displays a corresponding message.

7.3. Special requirements

The list of course sections must be paginated when it is large.

7.4. System state before the use-case starts

The student must be logged in.

7.5. System state after the use-case ends

None.

7.6. Extension points

None.

8. Use-case Register for course sections

8.1. Summary

This use-case allows a student to register for one or more open course sections.

8.2. Flow of events

Main flow

The use-case starts when the student selects the function to register for course sections.

1. The student selects the open course section(s) to register for.

2. The system checks the registration conditions: the section still has seats, the student has not already registered for it, and its schedule does not conflict with the sections already registered.
3. The system asks the student to confirm the registration.
4. If the student confirms, the system records the registration and reduces the remaining seats of the section.
5. The system reports the result of the registration.

Alternative flows

- The section is full: the system reports this and does not allow registration.
- Schedule conflict or prerequisites not met: the system displays an error message. The student can choose another section or stop registering.
- The student does not confirm: the system does not record the registration.

8.3. Special requirements

Registration must be handled correctly when many students register at the same time for a section with few seats left, so that the maximum capacity is never exceeded.

8.4. System state before the use-case starts

The student must be logged in.

8.5. System state after the use-case ends

If the use-case succeeds, the section is added to the student's list of registered sections. Otherwise, the system state does not change.

8.6. Extension points

None.

9. Use-case View registered sections

9.1. Summary

This use-case allows a student to view the list of course sections they have registered for.

9.2. Flow of events

Main flow

The use-case starts when the student selects the function to view registered sections.

1. The system displays the student's registered sections: section code, course name, teacher, schedule and registration date.
2. If the student chooses to cancel a registration, the use-case Cancel registration is performed.

Alternative flows

The student has not registered for any section: the system displays a corresponding message.

9.3. Special requirements

A student can only view their own sections.

9.4. System state before the use-case starts

The student must be logged in.

9.5. System state after the use-case ends

None.

9.6. Extension points

Use-case Cancel registration: if the student chooses to cancel a registration.

10. Use-case Cancel registration

10.1. Summary

This use-case allows a student to cancel the registration of a course section they have registered for.

10.2. Flow of events

Main flow

The use-case starts when the student chooses to cancel the registration of a section in the list of registered sections.

1. The system displays a dialog asking the student to confirm the cancellation.
2. If the student agrees, the system checks that the cancellation is valid (the section belongs to the student and the cancellation deadline has not passed).
3. The system cancels the registration and increases the remaining seats of the section.
4. The system reports the result of the cancellation.

Alternative flows

- The student does not agree: the system changes nothing and the use-case ends.
- The cancellation deadline has passed: the system displays an error message and does not allow the cancellation.

10.3. Special requirements

A student can only cancel their own sections.

10.4. System state before the use-case starts

The student must be logged in.

10.5. System state after the use-case ends

If the use-case succeeds, the section is removed from the student's registrations. Otherwise, the system state does not change.

10.6. Extension points

None.

11. Use-case View academic results

11.1. Summary

This use-case allows a student to view their own academic results.

11.2. Flow of events

Main flow

The use-case starts when the student selects the function to view academic results.

1. The system displays the results for each course section: section code, course name, credits, grade, and pass or fail result.
2. The system displays the student's grade point average and current academic status.

Alternative flows

There are no results yet: the system displays a corresponding message.

11.3. Special requirements

A student can only view their own results. Only grades that the teacher has locked are displayed.

11.4. System state before the use-case starts

The student must be logged in.

11.5. System state after the use-case ends

None.

11.6. Extension points

None.

12. Use-case View assigned student lists

12.1. Summary

This use-case allows a teacher to view the lists of students in the course sections they are assigned to teach.

12.2. Flow of events

Main flow

The use-case starts when the teacher selects the function to view assigned student lists.

1. The system displays the list of course sections the teacher is assigned to.
2. The teacher selects a course section.
3. The system displays the list of students in that section: student ID, full name and class.

Alternative flows

The teacher has not been assigned any course section: the system displays a corresponding message.

12.3. Special requirements

A teacher can only view the student lists of the sections they are assigned to. The list must be paginated.

12.4. System state before the use-case starts

The teacher must be logged in.

12.5. System state after the use-case ends

None.

12.6. Extension points

None.

13. Use-case Filter students

13.1. Summary

This use-case allows a teacher or an administrator to filter students by certain criteria. A teacher only sees students in the course sections assigned to them.

13.2. Flow of events

Main flow

The use-case starts when the user chooses the filter function.

1. The system displays the filter criteria (for example: class, academic cohort, major, academic status).
2. The user selects or enters the criteria and requests the filter.
3. The system displays the list of students that match the selected criteria.

Alternative flows

No student matches the criteria: the system displays a no-results message.

13.3. Special requirements

A teacher can only filter among students in the course sections they are assigned to.

13.4. System state before the use-case starts

The user (Teacher or Administrator) must be logged in.

13.5. System state after the use-case ends

None.

13.6. Extension points

None.

14. Use-case Open student detailed record

14.1. Summary

This use-case allows a teacher or an administrator to open the detailed record of a student selected from the search results. It extends the use-case Search students.

14.2. Flow of events

Main flow

The use-case starts when the user chooses to view the detailed record of a student in the search results.

1. If the user is a teacher, the system checks whether the student belongs to the teacher's assigned course sections.
2. The system displays the student's detailed record: personal information, contact details, current course sections, grades of the sections taught by the teacher, and academic status.

Alternative flows

The student is not in the teacher's assigned course sections: the system reports that the teacher has no permission to view the record and the use-case ends.

14.3. Special requirements

A teacher can only view the records of students in the sections they are responsible for, and cannot edit the record.

14.4. System state before the use-case starts

The user (Teacher or Administrator) must be logged in and the search results of the use-case Search students are displayed.

14.5. System state after the use-case ends

None.

14.6. Extension points

None.

15. Use-case Enter grades

15.1. Summary

This use-case allows a teacher to enter grades for the students of the course sections they are responsible for.

15.2. Flow of events

Main flow

The use-case starts when the teacher selects the function to enter grades for a course section.

1. The system displays the student list of the section and the grade columns to fill in.
2. The teacher enters grades for the students and requests to save.
3. The system performs the use-case Check valid grade range to check each entered grade.
4. If all grades are valid, the system saves the grades.
5. The system reports the result of the grade entry.

Alternative flows

- Some grades are outside the valid range: the system displays an error message and points out the invalid grades. The teacher can correct them or cancel the grade entry.
- The gradebook is locked: the system reports that grades cannot be entered and the use-case ends.

15.3. Special requirements

A teacher can only enter grades for the sections they are responsible for.

15.4. System state before the use-case starts

The teacher must be logged in and the gradebook of the section must not be locked.

15.5. System state after the use-case ends

If the use-case succeeds, the students' grades are saved in the system. Otherwise, the system state does not change.

15.6. Extension points

None. This use-case includes the use-case Check valid grade range.

16. Use-case Edit grades

16.1. Summary

This use-case allows a teacher to edit grades already entered for students, as long as the gradebook is not locked.

16.2. Flow of events

Main flow

The use-case starts when the teacher selects the function to edit grades of a course section.

1. The system displays the list of students with their current grades.
2. The teacher edits the grades of one or more students and requests to save.

3. The system performs the use-case Check valid grade range to check the edited grades.
4. If they are valid, the system updates the grades and records the edit history (editor, time, old grade, new grade).
5. The system reports the result of the update.

Alternative flows

- A grade is outside the valid range: the system displays an error message. The teacher can correct it or cancel the update.
- The gradebook is locked: the system reports that grades cannot be edited. The teacher must ask an administrator to unlock the gradebook.

16.3. Special requirements

Every grade change must be logged for auditing.

16.4. System state before the use-case starts

The teacher must be logged in and the gradebook of the section must not be locked.

16.5. System state after the use-case ends

If the use-case succeeds, the grades are updated and the edit history is saved. Otherwise, the system state does not change.

16.6. Extension points

None. This use-case includes the use-case Check valid grade range.

17. Use-case Check valid grade range

17.1. Summary

This use-case checks whether entered or edited grades are within the valid range of values. It is included by Enter grades and Edit grades.

17.2. Flow of events

Main flow

The use-case starts when the use-case Enter grades or Edit grades requests a grade check.

1. The system checks that each grade is a number within the valid range of the section's grading scale (for example from 0 to 10).
2. The system returns the check result to the calling use-case.

Alternative flows

Some grades are invalid (not a number or outside the allowed range): the system returns the list of invalid grades with an error message.

17.3. Special requirements

The valid range is taken from the configured grading scale of the section.

17.4. System state before the use-case starts

There are grades to check from the use-case Enter grades or Edit grades.

17.5. System state after the use-case ends

The system state does not change.

17.6. Extension points

None.

18. Use-case Lock gradebook

18.1. Summary

This use-case allows a teacher to lock the gradebook of a course section after finishing grade entry.

18.2. Flow of events

Main flow

The use-case starts when the teacher selects the function to lock the gradebook of a course section.

1. The system displays a warning that after locking, the teacher can no longer enter or edit grades, and asks for confirmation.

2. If the teacher confirms, the system checks that grades have been entered for all students of the section.
3. The system changes the gradebook to the "Locked" status and publishes the grades to the students.
4. The system reports the result of locking the gradebook.

Alternative flows

- The teacher does not confirm: the system changes nothing.
- Some students still have no grade: the system reports this and asks the teacher to enter all grades before locking.

18.3. Special requirements

A teacher can only lock the gradebook of the sections they are responsible for.

18.4. System state before the use-case starts

The teacher must be logged in and the gradebook must not be locked.

18.5. System state after the use-case ends

If the use-case succeeds, the gradebook is in the "Locked" status. Otherwise, the system state does not change.

18.6. Extension points

None.

19. Use-case Unlock gradebook

19.1. Summary

This use-case allows an administrator to unlock a locked gradebook so that the teacher can edit grades when needed.

19.2. Flow of events

Main flow

The use-case starts when the administrator selects the function to unlock a gradebook.

1. The system displays the list of locked gradebooks.

2. The administrator selects a gradebook and enters the reason for unlocking.
3. The system asks the administrator to confirm.
4. If the administrator confirms, the system changes the gradebook to the "Unlocked" status and records the history (who unlocked it, when, and why).
5. The system reports the result.

Alternative flows

- The administrator does not confirm: the system changes nothing.
- No reason has been entered: the system asks for a reason before unlocking.

19.3. Special requirements

Every unlock must be recorded in the history.

19.4. System state before the use-case starts

The administrator must be logged in.

19.5. System state after the use-case ends

If the use-case succeeds, the gradebook is unlocked and the teacher can edit grades. Otherwise, the system state does not change.

19.6. Extension points

None.

20. Use-case Create account

20.1. Summary

This use-case allows an administrator to create a login account for a student, a teacher or an administrator.

20.2. Flow of events

Main flow

The use-case starts when the administrator selects the function to create an account.

1. The system asks the administrator to enter the account information: full name, email, role (Student, Teacher, Administrator) and username.
2. The administrator enters the information and requests to create the account.
3. The system validates the information and creates the account with a randomly generated initial password.
4. The system reports the result of the account creation.

Alternative flows

- The username or email already exists: the system displays an error message. The administrator can correct it or cancel.
- Required information is missing: the system displays an error message and asks for the missing fields.

20.3. Special requirements

Passwords must be stored in encrypted form.

20.4. System state before the use-case starts

The administrator must be logged in.

20.5. System state after the use-case ends

If the use-case succeeds, the new account is added to the system. Otherwise, the system state does not change.

20.6. Extension points

None.

21. Use-case Edit account

21.1. Summary

This use-case allows an administrator to edit the information of an existing account.

21.2. Flow of events

Main flow

The use-case starts when the administrator selects the function to edit an account.

1. The system displays the information of the account to be edited.
2. The administrator edits the information (full name, email, role, password reset) as desired.
3. The administrator requests the update.
4. The system validates the information and updates it in the system.
5. The system reports the result of the update.

Alternative flows

- The administrator cancels: the system does not update the information.
- Invalid information or an email already used by another account: the system displays an error message. The administrator can correct it or cancel.

21.3. Special requirements

None.

21.4. System state before the use-case starts

The administrator must be logged in.

21.5. System state after the use-case ends

If the use-case succeeds, the account information is updated. Otherwise, the system state does not change.

21.6. Extension points

None.

22. Use-case Manage account status

22.1. Summary

This use-case allows an administrator to change the status of an account (activate, lock, deactivate).

22.2. Flow of events

Main flow

The use-case starts when the administrator selects the function to manage account status.

1. The system displays the list of accounts with their current status.
2. The administrator selects an account and chooses a new status.
3. The system asks the administrator to confirm.
4. If the administrator confirms, the system updates the account status.
5. The system reports the result.

Alternative flows

- The administrator does not confirm: the system changes nothing.
- The administrator tries to lock their own account: the system reports that this is not allowed.

22.3. Special requirements

A locked or deactivated account cannot log in to the system. The account list must be paginated.

22.4. System state before the use-case starts

The administrator must be logged in.

22.5. System state after the use-case ends

If the use-case succeeds, the account status is updated. Otherwise, the system state does not change.

22.6. Extension points

None.

23. Use-case Search students

23.1. Summary

This use-case allows a teacher or an administrator to search for students in the system.

23.2. Flow of events

Main flow

The use-case starts when the user selects the function to search for students.

1. The system displays a form for the user to enter search criteria (student ID, full name, class, major).
2. The user enters the criteria and requests the search.
3. The system searches and lists the students that match the criteria.
4. If the user selects a student, the use-case Open student detailed record is performed.

Alternative flows

No student matches the criteria: the system displays a no-results message.

23.3. Special requirements

Search results must be paginated when there are many. A teacher only sees students in the course sections assigned to them.

23.4. System state before the use-case starts

The user (Teacher or Administrator) must be logged in.

23.5. System state after the use-case ends

None.

23.6. Extension points

Use-case Open student detailed record: if the user chooses to view a student's detailed record.

24. Use-case Add student record

24.1. Summary

This use-case allows an administrator to add the record of a new student to the system.

24.2. Flow of events

Main flow

The use-case starts when the administrator selects the function to add a student record.

1. The system asks the administrator to enter the student information: student ID, full name, date of birth, gender, address, phone number, email, class and academic status.
2. The administrator enters the information and requests to add the record.
3. The system validates the information and adds the record to the system.
4. The system reports the result of the addition.

Alternative flows

- The student ID already exists: the system displays an error message. The administrator can correct it or cancel.
- Required information is missing or invalid: the system displays an error message and asks for the information again.

24.3. Special requirements

The student ID is unique in the system.

24.4. System state before the use-case starts

The administrator must be logged in.

24.5. System state after the use-case ends

If the use-case succeeds, the new student record is added to the system. Otherwise, the system state does not change.

24.6. Extension points

None.

25. Use-case Edit student information

25.1. Summary

This use-case allows an administrator to edit the information in a student's record.

25.2. Flow of events

Main flow

The use-case starts when the administrator selects the function to edit a student's information.

1. The system displays the student's current information.
2. The administrator edits the information as desired (except the student ID).
3. The administrator requests the update.
4. The system validates the information and updates it in the system.
5. The system reports the result of the update.

Alternative flows

- The administrator cancels: the system does not update the information.
- Invalid information or missing required fields: the system displays an error message. The administrator can correct it or cancel.

25.3. Special requirements

None.

25.4. System state before the use-case starts

The administrator must be logged in.

25.5. System state after the use-case ends

If the use-case succeeds, the student's information is updated. Otherwise, the system state does not change.

25.6. Extension points

None.

26. Use-case Edit academic status

26.1. Summary

This use-case allows an administrator to edit a student's academic status

26.2. Flow of events

Main flow

The use-case starts when the administrator selects the function to edit a student's academic status.

1. The system displays the student's current academic status.
2. The administrator selects the new academic status and enters a note (if any).
3. The administrator requests the update.
4. The system checks that the status change is valid, updates it in the system and records the change history.
5. The system reports the result of the update.

Alternative flows

- The administrator cancels: the system does not update the status.
- The status change is not valid: the system displays an error message. The administrator can choose again or cancel.

26.3. Special requirements

Every academic status change must be logged (editor, time, old status, new status).

26.4. System state before the use-case starts

The administrator must be logged in.

26.5. System state after the use-case ends

If the use-case succeeds, the student's academic status is updated. Otherwise, the system state does not change.

26.6. Extension points

None.

27. Use-case Manage courses

27.1. Summary

This use-case allows an administrator to manage (view, add, edit, delete) the courses in the system.

27.2. Flow of events

Main flow

The use-case starts when the administrator selects the function to manage courses.

1. The system displays the list of courses.
2. If the administrator chooses "Add", the Add course flow is performed.
3. If the administrator chooses "Update" for a course, the Update course flow is performed.
4. If the administrator chooses "Delete" for a course, the Delete course flow is performed.

Add course

1. The system asks the administrator to enter the course information (course code, course name, credits, description).
2. The administrator enters the information and requests to add the course.
3. The system validates the information and adds the course to the system.
4. The system reports the result of the addition.

Update course

1. The system displays the information of the course to be updated.
2. The administrator edits the information as desired and requests the update.
3. The system validates the information and updates it in the system.
4. The system reports the result of the update.

Delete course

1. The system asks the administrator to confirm the deletion of the course.
2. If the administrator confirms, the system checks that the course has no open course section and then deletes the course from the system.
3. The system reports the result of the deletion.

Alternative flows

- The course code already exists or the information is invalid: the system displays an error message. The administrator can correct it or cancel.
- The course already has open course sections or grades: the system reports that it cannot be deleted.

27.3. Special requirements

The course list must be paginated to improve system performance.

27.4. System state before the use-case starts

The administrator must be logged in.

27.5. System state after the use-case ends

If the use-case succeeds, the course information is added, updated or removed from the system. Otherwise, the system state does not change.

27.6. Extension points

None.

28. Use-case Create course sections

28.1. Summary

This use-case allows an administrator to create a new course section for a course so that students can register.

28.2. Flow of events

Main flow

The use-case starts when the administrator selects the function to create a course section.

1. The system asks the administrator to select a course and enter the section information: section code, semester, responsible teacher, schedule, classroom and maximum capacity.
2. The administrator enters the information and requests to create the section.
3. The system validates the information (no conflict with the teacher's or the classroom's schedule) and creates the section.
4. The system reports the result of creating the section.

Alternative flows

- The section code already exists: the system displays an error message. The administrator can correct it or cancel.
- The teacher or the classroom has a schedule conflict: the system displays an error message and asks the administrator to choose again.

28.3. Special requirements

None.

28.4. System state before the use-case starts

The administrator must be logged in and the course must already exist in the system.

28.5. System state after the use-case ends

If the use-case succeeds, the new section is added to the system and appears in the list of open course sections. Otherwise, the system state does not change.

28.6. Extension points

None.

29. Use-case View course section student list

29.1. Summary

This use-case allows an administrator to view the list of students registered for a course section.

29.2. Flow of events

Main flow

The use-case starts when the administrator selects the function to view the student list of a course section.

1. The system asks the administrator to select a course section.
2. The system displays the list of students registered for the section: student ID, full name, class and registration date.
3. The system displays the current enrollment and the maximum capacity of the section.

Alternative flows

No student has registered for the section: the system displays a corresponding message.

29.3. Special requirements

The student list must be paginated when it is large.

29.4. System state before the use-case starts

The administrator must be logged in.

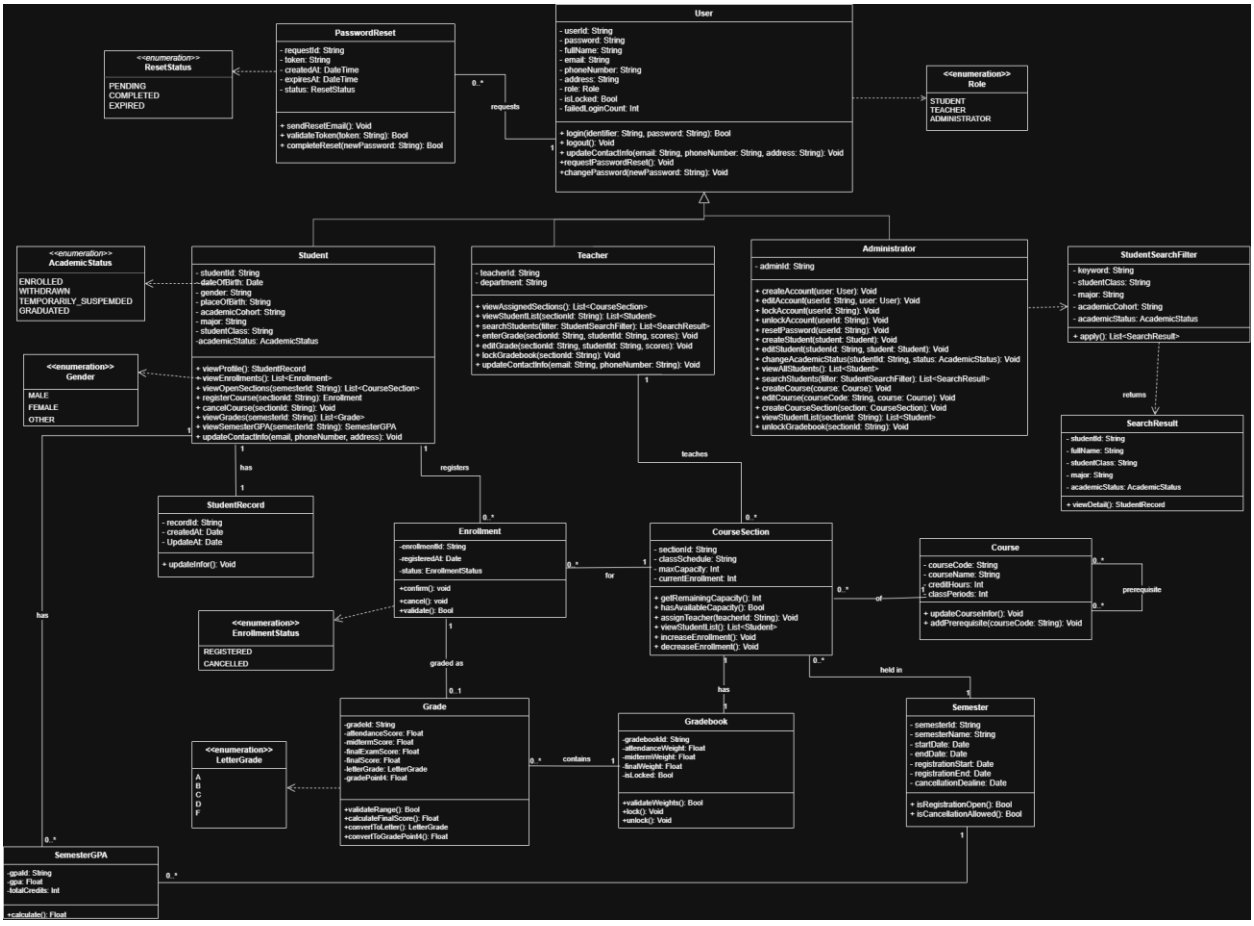
29.5. System state after the use-case ends

None.

29.6. Extension points

None.

IV. Class Diagram



1. Overview

1.1. List of Classes

No.	Class	Description
1	User	Abstract base class for everyone who logs in to the system (Student, Teacher, Administrator).
2	PasswordReset	Represents one password-reset request made by a user who forgot the password.
3	Student	A student of the university.
4	Teacher	A teacher/lecturer.
5	Administrator	The system administrator.
6	StudentRecord	The detailed record of a student (personal information plus history) shown when a record is opened from a list or search result.
7	Enrollment	A student's registration for one course section.
8	CourseSection	A class offering of a course in a semester, with schedule, capacity and assigned teacher.
9	Course	A subject in the curriculum, independent of semester.
10	Semester	A teaching term, with registration and cancellation periods.
11	Gradebook	The gradebook of one course section: grade-component weights and lock status.
12	Grade	The grades of one student's enrollment: component scores, final score, letter grade and 4-point grade.
13	SemesterGPA	The grade point average of a student for one semester.
14	StudentSearchFilter	Holds the search keyword and filter criteria used to search students.

15	SearchResult	One row of a student search/filter result.

1.2. List of Enumerations

No.	Enumeration	Description
1	Role	User account role
2	ResetStatus	Status of a password-reset request
3	AcademicStatus	Academic status of a student
4	Gender	Gender of a student
5	EnrollmentStatus	Status of a registration
6	LetterGrade	Letter grade

1.3. List of Relationships

No.	Relationship	Type	Multiplicity	Meaning
1	Student, Teacher, Administrator → User	Generalization	–	Specialised kinds of user
2	User – PasswordReset (requests)	Association	1 – 0..*	A user can make many reset requests
3	Student – StudentRecord (has)	Association	1 – 1	Each student has one detailed record
4	Student – Enrollment (registers)	Association	1 – 0..*	A student registers many sections
5	Enrollment – CourseSection (for)	Association	0..* – 1	A section has many enrollments

6	Enrollment – Grade (graded as)	Association	1 – 0..1	An enrollment has at most one grade
7	Teacher – CourseSection (teaches)	Association	1 – 0..*	A teacher teaches many sections
8	CourseSection – Course (of)	Association	0..* – 1	A course is offered as many sections
9	Course – Course (prerequisite)	Reflexive association	0..* – 0..*	A course may require other courses
10	CourseSection – Semester (held in)	Association	0..* – 1	Sections are held in a semester
11	CourseSection – Gradebook (has)	Association	1 – 1	Each section has one gradebook
12	Gradebook – Grade (contains)	Association	1 – 0..*	A gradebook contains many grades
13	Student – SemesterGPA (has)	Association	1 – 0..*	One GPA per semester per student
14	Semester – SemesterGPA	Association	1 – 0..*	GPA belongs to a semester
15	StudentSearchFilter → SearchResult (returns)	Dependency	–	Filter produces result rows
16	Administrator, Teacher → StudentSearchFilter	Dependency	–	Use the filter to search students
17	Classes → enumerations	Dependency	–	Attribute types (Role, AcademicStatus, ...)

2. Class User

2.1. Summary

Abstract base class for everyone who logs in to the system (Student, Teacher, Administrator). It holds the common account data and the common behaviours: login, logout, contact update and password handling.

2.2. Attributes

Attribute	Type	Description
userId	String	Unique identifier of the account (used as username)
password	String	Password, stored in encrypted (hashed) form
fullName	String	Full name of the user
email	String	Email address (must be unique)
phoneNumber	String	Phone number
address	String	Contact address
role	Role	Role of the user: STUDENT, TEACHER or ADMINISTRATOR
isLocked	Bool	True if the account is locked/deactivated and cannot log in
failedLoginCount	Int	Number of consecutive failed login attempts

2.3. Methods

Method	Description
login(identifier: String, password: String): Bool	Checks the identifier (student ID; teacher ID or email) and password; refuses locked accounts; returns true on success.

logout(): Void	Ends the current session.
updateContactInfo(email, phoneNumber, address): Void	Validates and updates the user's own contact details.
requestPasswordReset(): Void	Creates a PasswordReset request for this user.
changePassword(newPassword: String): Void	Replaces the password with a new one (stored encrypted).

2.4. Relationships

- Generalization: Student, Teacher and Administrator inherit from User.
- 1 User – 0..* PasswordReset ("requests").
- Dependency on Role (attribute role).

2.5. Constraints and business rules

- Passwords must be stored in encrypted form (UC Log in, Create account).
- A locked or deactivated account (isLocked = true) cannot log in (UC Manage account status).
- A user can only use the functions and see the data permitted by his/her role.

3. Class PasswordReset

3.1. Summary

Represents one password-reset request made by a user who forgot the password.

3.2. Attributes

Attribute	Type	Description
requestId	String	Unique identifier of the request
token	String	Randomly generated reset token sent to the user's email
createdAt	DateTime	Time the request was created

expiresAt	DateTime	Time after which the token is no longer valid
status	ResetStatus	PENDING, COMPLETED or EXPIRED

3.3. Methods

Method	**Description**
sendResetEmail(): Void	Sends the reset instructions/new password to the user's registered email.
validateToken(token: String): Bool	Checks that the token matches, has not expired and the request is still PENDING.
completeReset(newPassword: String): Bool	Sets the new (encrypted) password and changes the status to COMPLETED.

3.4. Relationships

- Association: 0..* PasswordReset – 1 User ("requests").
- Dependency on ResetStatus.

3.5. Constraints and business rules

- The token/new password must be randomly generated.
- Expired or completed requests cannot be used again.
- If the entered ID/email does not exist, no request is created (UC Request password reset).

4. Class Student

4.1. Summary

A student of the university. Extends User with academic data and the functions a student can perform: viewing profile, registering/cancelling course sections and viewing results.

4.2. Attributes

Attribute	Type	Description
studentId	String	Unique student ID (also the login name)
dateOfBirth	Date	Date of birth
gender	Gender	MALE, FEMALE or OTHER
placeOfBirth	String	Place of birth
academicCohort	String	Academic cohort (intake year)
major	String	Major
studentClass	String	Student class
academicStatus	AcademicStatus	ENROLLED, WITHDRAWN, TEMPORARILY_SUSPENDED or GRADUATED

4.3. Methods

Method	Description
viewProfile(): StudentRecord	Shows the student's own personal information (UC View personal information).
viewEnrollments(): List<Enrollment>	Lists the student's registered sections (UC View registered sections).
viewOpenSections(semesterId: String): List<CourseSection>	Lists sections open for registration, with remaining seats (UC View open course sections).
registerCourse(sectionId: String): Enrollment	Registers for a section after the checks of Enrollment.validate() (UC Register for course sections).

cancelCourse(sectionId: String): Void	Cancels a registration before the deadline (UC Cancel registration).
viewGrades(semesterId: String): List<Grade>	Shows own component/final grades of locked gradebooks (UC View academic results).
viewSemesterGPA(semesterId: String): SemesterGPA	Shows own semester GPA.
updateContactInfo(email, phoneNumber, address): Void	Updates only contact details (UC Update contact details).

4.4. Relationships

- Generalization: Student is a subclass of User.
- 1 Student – 1 StudentRecord ("has").
- 1 Student – 0..* Enrollment ("registers").
- 1 Student – 0..* SemesterGPA ("has").
- Dependencies on Gender and AcademicStatus.

4.5. Constraints and business rules

- studentId is unique and cannot be edited after creation.
- A student can only view/update his/her own data and can change only contact details; other information is changed by the Administrator.
- Only grades of locked gradebooks are visible to the student.

5. Class Teacher

5.1. Summary

A teacher/lecturer. Extends User with the functions to view the sections he/she teaches, search students within them, and enter, edit and lock grades.

5.2. Attributes

Attribute	Type	Description

teacherId	String	Unique teacher ID (can be used to log in)
department	String	Department the teacher belongs to

5.3. Methods

Method	**Description**
viewAssignedSections(): List<CourseSection>	Lists the sections assigned to the teacher.
viewStudentList(sectionId: String): List<Student>	Lists students of an assigned section (UC View assigned student lists).
searchStudents(filter: StudentSearchFilter): List<SearchResult>	Searches/filters students, limited to assigned sections (UC Filter students).
enterGrade(sectionId, studentId: String, scores): Void	Enters component scores for a student (UC Enter grades).
editGrade(sectionId, studentId: String, scores): Void	Edits entered scores while the gradebook is unlocked (UC Edit grades).
lockGradebook(sectionId: String): Void	Locks the gradebook after all grades are entered (UC Lock gradebook).
updateContactInfo(email, phoneNumber): Void	Updates the teacher's phone number and email.

5.4. Relationships

- Generalization: Teacher is a subclass of User.
- 1 Teacher – 0..* CourseSection ("teaches").

5.5. Constraints and business rules

- A teacher can only see students and grades of the sections he/she is assigned to.
- A teacher can enter/edit grades only while the gradebook is not locked.

- A teacher cannot unlock a gradebook; only the Administrator can.

6. Class Administrator

6.1. Summary

The system administrator. Extends User with management functions for accounts, student records, courses, course sections and gradebooks.

6.2. Attributes

Attribute	Type	Description
adminId	String	Unique administrator ID

6.3. Methods

Method	Description
createAccount(user: User): Void	Creates an account with a random initial password (UC Create account).
editAccount(userId: String, user: User): Void	Edits account information (UC Edit account).
lockAccount(userId: String): Void / unlockAccount(userId: String): Void	Locks or unlocks an account (UC Manage account status).
resetPassword(userId: String): Void	Resets a user's password on request.
createStudent(student: Student): Void	Adds a student record (UC Add student record).
editStudent(studentId: String, student: Student): Void	Edits student information (UC Edit student information).
changeAcademicStatus(studentId: String, status: AcademicStatus): Void	Changes academic status and logs the change (UC Edit academic status).

viewAllStudents(): List<Student>	Lists all students.
searchStudents(filter: StudentSearchFilter): List<SearchResult>	Searches and filters all students (UC Search students).
createCourse(course) / editCourse(courseCode, course): Void	Adds or edits courses (UC Manage courses).
createCourseSection(section: CourseSection): Void	Creates a section for a semester (UC Create course sections).
viewStudentList(sectionId: String): List<Student>	Views students of any section (UC View course section student list).
unlockGradebook(sectionId: String): Void	Unlocks a locked gradebook with a reason (UC Unlock gradebook).

6.4. Relationships

- Generalization: Administrator is a subclass of User.
- Dependency (uses) on StudentSearchFilter.

6.5. Constraints and business rules

- An administrator cannot lock his/her own account.
- Every unlock of a gradebook and every academic status change must be logged (who, when, why).

7. Class StudentRecord

7.1. Summary

The detailed record of a student (personal information plus history) shown when a record is opened from a list or search result.

7.2. Attributes

Attribute	Type	Description
recordId	String	Unique identifier of the record

createdAt	Date	Date the record was created
updatedAt	Date	Date of the last update

7.3. Methods

Method	Description
updateInfo(): Void	Updates the record and refreshes updatedAt.

7.4. Relationships

- 1 StudentRecord – 1 Student ("has").
- Returned by Student.viewProfile() and SearchResult.viewDetail().

7.5. Constraints and business rules

- Teachers can open a record only for students of their sections and cannot edit it.
- A record is changed only by the Administrator (the student may change contact details only).

8. Class Enrollment

8.1. Summary

A student's registration for one course section. It is the link class between Student and CourseSection and the anchor of the grade.

8.2. Attributes

Attribute	Type	Description
enrollmentId	String	Unique identifier
registeredAt	Date	Date of registration
status	EnrollmentStatus	REGISTERED or CANCELLED

8.3. Methods

Method	Description
confirm(): Void	Confirms the registration and records it; increases section enrollment.
cancel(): Void	Cancels the registration; decreases section enrollment.
validate(): Bool	Checks registration period, remaining capacity, duplicate registration, schedule conflict and prerequisites; returns the reason when invalid.

8.4. Relationships

- Student 1 – 0..* Enrollment ("registers").
- Enrollment 0..* – 1 CourseSection ("for").
- Enrollment 1 – 0..1 Grade ("graded as").
- Dependency on EnrollmentStatus.

8.5. Constraints and business rules

- A student cannot have two REGISTERED enrollments for the same section.
- Cancellation is allowed only before the cancellation deadline of the semester.
- Concurrent registrations must never make the number of students exceed the maximum capacity.

9. Class CourseSection

9.1. Summary

A class offering of a course in a semester, with schedule, capacity and assigned teacher.

9.2. Attributes

Attribute	Type	Description

sectionId	String	Unique section code
classSchedule	String	Class schedule (days, periods, classroom)
maxCapacity	Int	Maximum number of students
currentEnrollment	Int	Current number of registered students

9.3. Methods

Method	**Description**
getRemainingCapacity(): Int	Returns maxCapacity − currentEnrollment.
hasAvailableCapacity(): Bool	True if at least one seat remains.
assignTeacher(teacherId: String): Void	Assigns the responsible teacher (no schedule conflict).
viewStudentList(): List<Student>	Returns the registered students.
increaseEnrollment(): Void	Adds one registered student.
decreaseEnrollment(): Void	Removes one registered student.

9.4. Relationships

- Teacher 1 – 0..* CourseSection ("teaches").
- CourseSection 0..* – 1 Course ("of").
- CourseSection 0..* – 1 Semester ("held in").
- CourseSection 1 – 1 Gradebook ("has").
- Enrollment 0..* – 1 CourseSection ("for").

9.5. Constraints and business rules

- $0 \leq$ currentEnrollment $\leq$ maxCapacity at all times.
- Section code is unique.

- Teacher and classroom must have no schedule conflict (UC Create course sections).

10. Class Course

10.1. Summary

A subject in the curriculum, independent of semester.

10.2. Attributes

Attribute	Type	Description
courseCode	String	Unique course code
courseName	String	Course name
creditHours	Int	Credit hours
classPeriods	Int	Number of class periods

10.3. Methods

Method	Description
updateCourseInfo(): Void	Validates and updates the course information.
addPrerequisite(courseCode: String): Void	Adds another course as a prerequisite.

10.4. Relationships

- 1 Course – 0..* CourseSection ("of").
- Reflexive association 0..* – 0..* ("prerequisite").

10.5. Constraints and business rules

- courseCode is unique.
- A course that has course sections or grades cannot be deleted.
- A course cannot be its own (direct or indirect) prerequisite.

11. Class Semester

11.1. Summary

A teaching term, with registration and cancellation periods.

11.2. Attributes

Attribute	Type	Description
semesterId	String	Unique identifier
semesterName	String	Name, e.g. 2026-2027 / Semester 1
startDate	Date	Start date
endDate	Date	End date
registrationStart	Date	Start of registration period
registrationEnd	Date	End of registration period
cancellationDeadline	Date	Last day to cancel a registration

11.3. Methods

Method	Description
isRegistrationOpen(): Bool	True if today is within the registration period.
isCancellationAllowed(): Bool	True if today is not after the cancellation deadline.

11.4. Relationships

- 1 Semester – 0..* CourseSection ("held in").
- 1 Semester – 0..* SemesterGPA.

11.5. Constraints and business rules

- registrationStart ≤ registrationEnd ≤ cancellationDeadline ≤ endDate (recommended).

12. Class Gradebook

12.1. Summary

The gradebook of one course section: grade-component weights and lock status.

12.2. Attributes

Attribute	Type	Description
gradebookId	String	Unique identifier
attendanceWeight	Float	Weight of attendance
midtermWeight	Float	Weight of midterm exam
finalWeight	Float	Weight of final exam
isLocked	Bool	True when locked

12.3. Methods

Method	Description
validateWeights(): Bool	Checks that the three weights sum to 100%.
lock(): Void	Locks the gradebook and publishes grades to students.
unlock(): Void	Unlocks it (Administrator only) and records the history.

12.4. Relationships

- 1 CourseSection – 1 Gradebook ("has").
- 1 Gradebook – 0..* Grade ("contains").

12.5. Constraints and business rules

- attendanceWeight + midtermWeight + finalWeight = 100%.
- Cannot lock while any student has no grade.
- While locked, grades cannot be entered or edited.

13. Class Grade

13.1. Summary

The grades of one student's enrollment: component scores, final score, letter grade and 4-point grade.

13.2. Attributes

Attribute	Type	Description
gradeId	String	Unique identifier
attendanceScore	Float	Attendance score
midtermScore	Float	Midterm score
finalExamScore	Float	Final examination score
finalScore	Float	Final course grade (weighted)
letterGrade	LetterGrade	A, B, C, D or F
gradePoint4	Float	Grade on the four-point scale

13.3. Methods

Method	Description
validateRange(): Bool	Checks each score is a number within the valid range (e.g. 0–10) (UC Check valid grade range).
calculateFinalScore(): Float	Weighted sum using the gradebook weights.

convertToLetter(): LetterGrade	Converts the final score into a letter grade.
convertToGradePoint4(): Float	Converts the final score into the four-point scale.

13.4. Relationships

- 1 Gradebook – 0..* Grade ("contains").
- 1 Enrollment – 0..1 Grade ("graded as").
- Dependency on LetterGrade.

13.5. Constraints and business rules

- Scores must be valid before saving; invalid grades are rejected with the list of errors.
- Every edit is logged (editor, time, old and new grade).
- Visible to the student only after the gradebook is locked.

14. Class SemesterGPA

14.1. Summary

The grade point average of a student for one semester.

14.2. Attributes

Attribute	Type	Description
gpaId	String	Unique identifier
gpa	Float	Semester GPA
totalCredits	Int	Total credits counted

14.3. Methods

Method	Description
calculate(): Float	Average of four-point grades weighted by credit hours.

14.4. Relationships

- 1 Student – 0..* SemesterGPA ("has").
- 1 Semester – 0..* SemesterGPA.

14.5. Constraints and business rules

- GPA = Σ(gradePoint4 × creditHours) / Σ(creditHours) over the student's graded sections of the semester.

15. Class StudentSearchFilter

15.1. Summary

Holds the search keyword and filter criteria used to search students.

15.2. Attributes

Attribute	Type	Description
keyword	String	Student ID or full name (partial or complete)
studentClass	String	Student class
major	String	Major
academicCohort	String	Academic cohort
academicStatus	AcademicStatus	Academic status

15.3. Methods

Method	Description
apply(): List<SearchResult>	Runs the search with the criteria and returns matching students.

15.4. Relationships

- Used by Administrator and Teacher (dependency).
- Dependency on SearchResult ("returns") and AcademicStatus.

15.5. Constraints and business rules

- Teachers' results are restricted to students in their assigned sections.
- An empty result produces a no-results notification.

16. Class SearchResult

16.1. Summary

One row of a student search/filter result.

16.2. Attributes

Attribute	Type	Description
studentId	String	Student ID
fullName	String	Full name
studentClass	String	Student class
major	String	Major
academicStatus	AcademicStatus	Academic status

16.3. Methods

Method	Description
viewDetail(): StudentRecord	Opens the detailed record of the student (UC Open student detailed record).

16.4. Relationships

- Returned by StudentSearchFilter.apply().

16.5. Constraints and business rules

- Results are paginated when many.

17. Enumerations

17.1. Role

User account role. Values: STUDENT, TEACHER, ADMINISTRATOR.

17.2. ResetStatus

Status of a password-reset request. Values: PENDING, COMPLETED, EXPIRED.

17.3. AcademicStatus

Academic status of a student. Values: ENROLLED, WITHDRAWN, TEMPORARILY_SUSPENDED, GRADUATED.

17.4. Gender

Gender of a student. Values: MALE, FEMALE, OTHER.

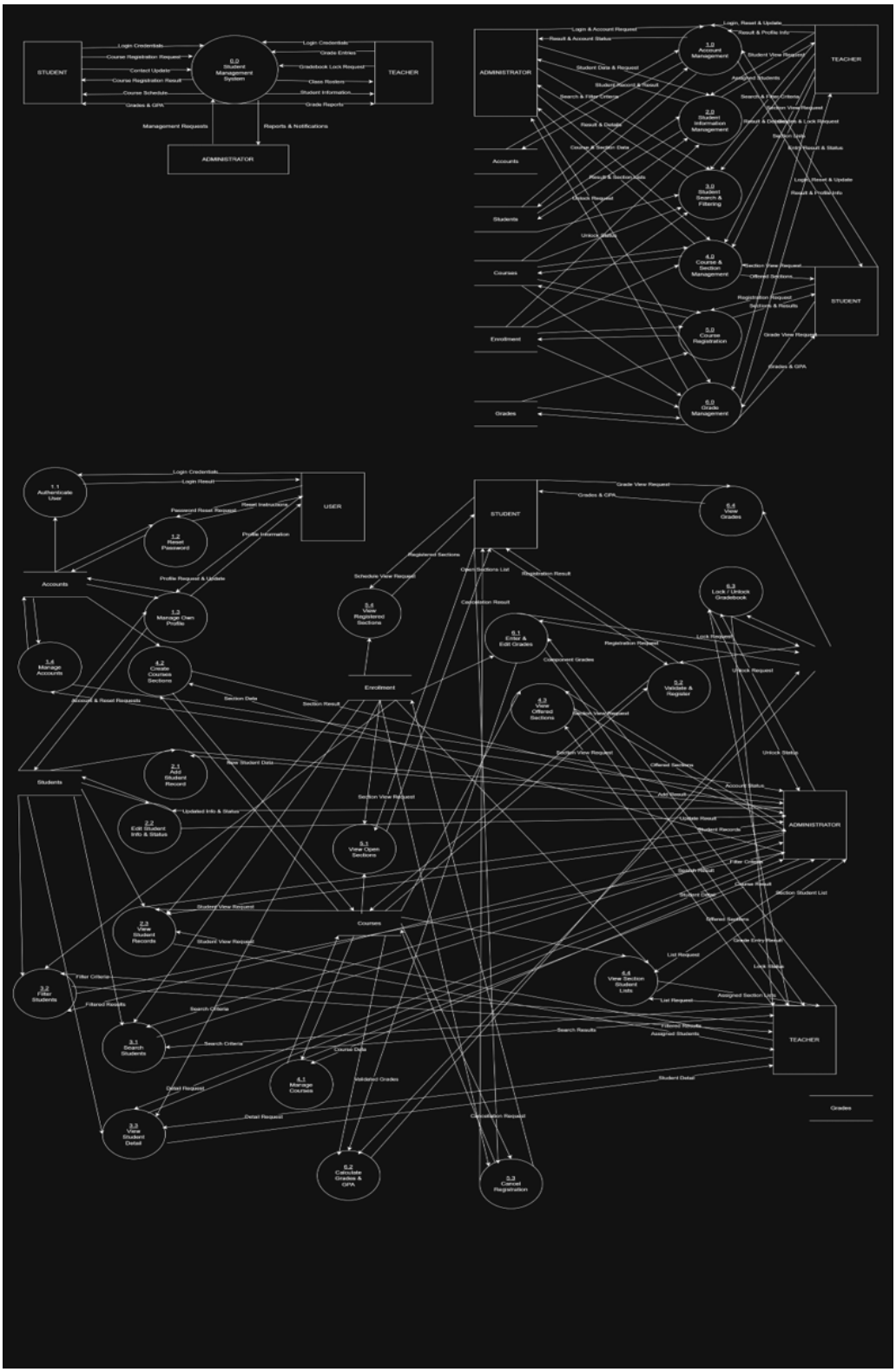
17.5. EnrollmentStatus

Status of a registration. Values: REGISTERED, CANCELLED.

17.6. LetterGrade

Letter grade. Values: A, B, C, D, F.

V. DFD Diagram



1. Overview

The DFD is organised in three levels: Level 0 (context diagram, process 0.0), Level 1 (processes 1.0–6.0) and Level 2 (decomposition of each Level 1 process into sub-processes x.1, x.2, ...).

1.1. List of External Entities

No.	Entity	Description
1	Student	Logs in, updates contact details, views and registers/cancels course sections, views grades and GPA.
2	Teacher	Logs in, views assigned students and sections, searches/filters students, enters/edits grades, locks the gradebook.
3	Administrator	Manages accounts, students, courses, sections, and unlocks gradebooks.
4	User	Generic entity in the Level 2 diagram of process 1.0 representing any Student, Teacher or Administrator who logs in, resets a password or manages a profile.

1.2. List of Data Stores

No.	Data store	Content
1	Accounts	Login accounts and password-reset requests.
2	Students	Student personal and academic data, student records.
3	Courses	Courses, course sections, semesters, schedules, capacity, teachers, prerequisites.
4	Enrollment	Registrations of students in course sections.
5	Grades	Gradebooks, component/final grades, letter grades, semester GPA, lock status.

1.3. List of Processes (Level 1)

No.	Process	Description	Data stores used
1.0	Account Management	Authentication, password reset, own-profile management and account administration.	Accounts, Students
2.0	Student Information Management	Adding, editing and viewing student records and academic status.	Students, Courses, Enrollment
3.0	Student Search & Filtering	Searching, filtering and opening student detail for the Administrator and Teachers.	Students, Enrollment
4.0	Course & Section Management	Managing courses, creating sections, viewing offered sections and section student lists.	Accounts, Courses, Enrollment
5.0	Course Registration	Viewing open sections, registering, cancelling and viewing registered sections.	Courses, Enrollment, Grades
6.0	Grade Management	Entering/editing grades, calculating results and GPA, locking/unlocking gradebooks, viewing grades.	Enrollment, Courses, Grades

1.4. List of Sub-processes (Level 2)

No.	Sub-process	Description	Data stores
1.1	Authenticate User	Verifies the identity of a user and starts a session.	Accounts
1.2	Reset Password	Handles a forgotten-password request.	Accounts

1.3	Manage Own Profile	Lets a user view their profile and update permitted contact data.	Accounts; Students
1.4	Manage Accounts	Administrator creates, edits, locks and unlocks accounts and handles reset requests.	Accounts
2.1	Add Student Record	Adds a new student record.	Students
2.2	Edit Student Info & Status	Edits student information and academic status.	Students
2.3	View Student Records	Shows student records to authorised users.	Students; Courses; Enrollment
3.1	Search Students	Searches students by keyword.	Students; Enrollment
3.2	Filter Students	Filters students by criteria.	Students; Enrollment
3.3	View Student Detail	Opens the detailed record of a student from the results.	Students; Enrollment
4.1	Manage Courses	Adds, edits and deletes courses.	Courses
4.2	Create Course Sections	Creates a course section for a semester.	Courses; Accounts
4.3	View Offered Sections	Lists the sections offered in a semester.	Courses
4.4	View Section Student Lists	Shows the students registered in a section.	Courses; Enrollment
5.1	View Open Sections	Shows sections open for registration.	Courses; Enrollment
5.2	Validate & Register	Validates and records a registration.	Courses; Enrollment; Grades
5.3	Cancel Registration	Cancels an existing registration.	Courses; Enrollment

5.4	View Registered Sections	Shows the student's registered sections and schedule.	Enrollment
6.1	Enter & Edit Grades	Teacher enters or edits component grades.	Enrollment; Grades
6.2	Calculate Grades & GPA	Computes final grade, letter grade, four-point grade and semester GPA.	Courses; Grades
6.3	Lock / Unlock Gradebook	Locks a gradebook (Teacher) or unlocks it (Administrator).	Grades
6.4	View Grades	Student views own results.	Grades

2. Level 0 – Context Diagram

2.1. Summary

The whole system is shown as one process, 0.0 Student Management System, which exchanges data with three external entities: Student, Teacher and Administrator.

2.2. Data flows

No.	Data flow	From	To	Description
1	Login Credentials	Student	System	Student ID and password to log in.
2	Course Registration Request	Student	System	Request to register for a course section.
3	Contact Update	Student	System	New email, phone number or address of the student.
4	Course Registration Result	System	Student	Accepted/rejected result of a registration or cancellation, with reason.

5	Course Schedule	System	Student	Registered sections and schedule of the student.
6	Grades & GPA	System	Student	Component grades, final grades, letter grades and semester GPA.
7	Login Credentials	Teacher	System	Teacher ID or email and password to log in.
8	Grade Entries	Teacher	System	Component grades entered or edited for students.
9	Gradebook Lock Request	Teacher	System	Request to lock the gradebook of a section.
10	Class Rosters	System	Teacher	Student lists of the assigned course sections.
11	Student Information	System	Teacher	Information/detailed record of students in assigned sections.
12	Grade Reports	System	Teacher	Grade entry result and gradebook status.
13	Management Requests	Administrator	System	Requests to manage accounts, students, courses, sections and gradebook unlocking.
14	Reports & Notifications	System	Administrator	Results, lists, reports and notifications returned to the administrator.

3. Level 1 Diagram

3.1. Summary

Process 0.0 is decomposed into six processes: Account Management, Student Information Management, Student Search & Filtering, Course & Section Management, Course Registration and Grade Management. They share five data stores: Accounts, Students, Courses, Enrollment and Grades.

3.2. Processes

Process	Inputs	Outputs	Data stores
1.0 Account Management	Login & Account Request; Login, Reset & Update	Result & Account Status; Result & Profile Info	Accounts, Students
2.0 Student Information Management	Student Data & Request; Student View Request	Student Record & Result; Assigned Students	Students, Courses, Enrollment
3.0 Student Search & Filtering	Search & Filter Criteria	Result & Details	Students, Enrollment
4.0 Course & Section Management	Course & Section Data; Section View Request	Result & Section Lists; Section Lists; Offered Sections	Accounts, Courses, Enrollment
5.0 Course Registration	Registration Request	Sections & Results	Courses, Enrollment, Grades
6.0 Grade Management	Grades & Lock Request; Unlock Request; Grade View Request	Entry Result & Status; Unlock Status; Grades & GPA	Enrollment, Courses, Grades

3.3. Data flows

No.	Data flow	From	To	Description

1	Login & Account Request	Administrator	1.0	Login and account-management requests.
2	Result & Account Status	1.0	Administrator	Login result and account status.
3	Login, Reset & Update	Teacher / Student	1.0	Login credentials, password-reset request and profile/contact update.
4	Result & Profile Info	1.0	Teacher / Student	Login result, reset instructions and profile information.
5	Student Data & Request	Administrator	2.0	New/updated student data, academic status and view requests.
6	Student Record & Result	2.0	Administrator	Student records and operation results.
7	Student View Request	Teacher	2.0	Request to view students.
8	Assigned Students	2.0	Teacher	Students of the teacher's sections.
9	Search & Filter Criteria	Administrator	3.0	Search keyword and filter criteria.
10	Search & Filter Criteria	Teacher	3.0	Search keyword and filter criteria (assigned students only).
11	Result & Details	3.0	Administrator / Teacher	Search/filter results and student detail.
12	Course & Section Data	Administrator	4.0	Course and section data to create/update.

13	Result & Section Lists	4.0	Administrator	Operation result, offered sections and section student lists.
14	Section View Request	Teacher	4.0	Request to view offered sections or assigned section lists.
15	Section Lists	4.0	Teacher	Offered sections and student lists of assigned sections.
16	Section View Request	Student	4.0	Request to view offered sections.
17	Offered Sections	4.0	Student	List of sections offered in the semester.
18	Registration Request	Student	5.0	Open-section, registration, cancellation and schedule requests.
19	Sections & Results	5.0	Student	Open sections, registered sections and registration/cancellation results.
20	Grades & Lock Request	Teacher	6.0	Grade entries/edits and gradebook lock request.
21	Entry Result & Status	6.0	Teacher	Grade entry result and lock status.
22	Unlock Request	Administrator	6.0	Request (with reason) to unlock a gradebook.
23	Unlock Status	6.0	Administrator	Result of the unlock.
24	Grade View Request	Student	6.0	Request to view own grades.
25	Grades & GPA	6.0	Student	Own grades and semester GPA.

4. Level 2 – Process 1.0 Account Management

4.1. Summary

Authentication, password reset, own-profile management and account administration.

4.2. Sub-processes and data stores

No.	Sub-process	Description	Data stores
1.1	Authenticate User	Verifies the identity of a user and starts a session.	Accounts
1.2	Reset Password	Handles a forgotten-password request.	Accounts
1.3	Manage Own Profile	Lets a user view their profile and update permitted contact data.	Accounts; Students
1.4	Manage Accounts	Administrator creates, edits, locks and unlocks accounts and handles reset requests.	Accounts

4.3. Data flows

No.	Data flow	From	To	Description
1	Login Credentials	User	1.1	Identifier (student ID; teacher ID or email) and password.
2	Login Result	1.1	User	Success with role-based functions, or an error (wrong credentials / locked account).
3	Password Reset Request	User	1.2	Registered email or ID of the user who forgot the password.

4	Reset Instructions	1.2	User	Instructions or new random password sent to the registered email, and the result.
5	Profile Request & Update	User	1.3	Request to view the profile and new contact data to update.
6	Profile Information	1.3	User	The user's own profile information and update result.
7	Account & Reset Requests	Administrator	1.4	Create, edit, lock, unlock or reset-password request for an account.
8	Account Status	1.4	Administrator	Result of the request and the current account status.

4.4. Processing description

1.1 Authenticate User

1. Receive the identifier (student ID; teacher ID or email) and password.

2. Look up the account and compare the encrypted password.

3. Check that the account is not locked or inactive.

4. Return the login result and the functions matching the user's role.

1.2 Reset Password

1. Receive the registered email or ID.

2. Check that the account exists.

3. Create a reset request/random new password, store it encrypted, and send the instructions to the registered email.

4. Return the result.

1.3 Manage Own Profile

1. Return the user's own information.

2. On update, validate the new contact data.

3. Save the changes and report the result.

1.4 Manage Accounts

1. Receive the account request (create, edit, lock, unlock, reset password).

2. Validate (unique username/email, required fields).

3. Create the account with a random initial password or update it.

4. Return the account status.

4.5. Business rules

1.1 Authenticate User

- Passwords are stored encrypted.
- Locked/deactivated accounts cannot log in.
- On wrong credentials an error message is returned and nothing changes.

1.2 Reset Password

- New password is randomly generated and stored encrypted.
- Unknown email/ID produces an error message and no change.
- A user may alternatively ask the Administrator (flow 1.4).

1.3 Manage Own Profile

- Students may change only email, phone number and address.
- Teachers may change only phone number and email.
- Other data is changed only by the Administrator.
- Invalid data is rejected and the existing data is kept.

1.4 Manage Accounts

- Only the Administrator can manage accounts.
- Administrator cannot lock his/her own account.
- Account lists are paginated.

5. Level 2 – Process 2.0 Student Information Management

5.1. Summary

Adding, editing and viewing student records and academic status.

5.2. Sub-processes and data stores

No.	Sub-process	Description	Data stores
2.1	Add Student Record	Adds a new student record.	Students
2.2	Edit Student Info & Status	Edits student information and academic status.	Students
2.3	View Student Records	Shows student records to authorised users.	Students; Courses; Enrollment

5.3. Data flows

No.	Data flow	From	To	Description
1	New Student Data	Administrator	2.1	Student ID, name, date of birth, gender, contact data, place of birth, cohort, major, class.
2	Add Result	2.1	Administrator	Success or error message (e.g. duplicate student ID).
3	Updated Info & Status	Administrator	2.2	Edited student information and/or

				new academic status with note.
4	Update Result	2.2	Administrator	Success or error message of the update.
5	Student View Request	Administrator, Teacher	2.3	Request to view student records.
6	Student Records	2.3	Administrator	All student records (Administrator).
7	Assigned Students	2.3	Teacher	Students of the teacher's assigned sections.

5.4. Processing description

2.1 Add Student Record

1. Receive student ID, full name, date of birth, gender, contact data, place of birth, cohort, major, class.

2. Validate required fields and uniqueness of the student ID.

3. Store the record.

4. Return the result.

2.2 Edit Student Info & Status

1. Read the current record.

2. Validate the updated data / the status change (Enrolled, Withdrawn, Temporarily Suspended, Graduated).

3. Save the changes and log the status change (editor, time, old/new status).

4. Return the result.

2.3 View Student Records

1. Receive the view request.

2. For the Administrator, read all students.

3. For a Teacher, use Enrollment and Courses to find the teacher's sections and return only their students.

4. Return the records.

5.5. Business rules

2.1 Add Student Record

- Student ID is unique.
- Missing/invalid data produces an error message.

2.2 Edit Student Info & Status

- Student ID cannot be changed.
- If the update fails, existing data is not modified.

2.3 View Student Records

- Administrator sees all students; Teacher only students of assigned sections.
- Teachers cannot edit records.

6. Level 2 – Process 3.0 Student Search & Filtering

6.1. Summary

Searching, filtering and opening student detail for the Administrator and Teachers.

6.2. Sub-processes and data stores

No.	Sub-process	Description	Data stores
3.1	Search Students	Searches students by keyword.	Students; Enrollment
3.2	Filter Students	Filters students by criteria.	Students; Enrollment
3.3	View Student Detail	Opens the detailed record of a student from the results.	Students; Enrollment

6.3. Data flows

No.	Data flow	From	To	Description
1	Search Criteria	Administrator, Teacher	3.1	Student ID, full name (partial or complete), class or major.
2	Search Results	3.1	Administrator, Teacher	Student ID, full name, class, major, academic status; or a no-result notice.
3	Filter Criteria	Administrator, Teacher	3.2	Class, academic cohort, major and academic status.
4	Filtered Results	3.2	Administrator, Teacher	Filtered list of students; or a no-result notice.
5	Detail Request	Administrator, Teacher	3.3	Selection of one student from the results.
6	Student Detail	3.3	Administrator, Teacher	Detailed record: personal and contact data, sections, academic status.

6.4. Processing description

3.1 Search Students

1. Receive student ID, full name (partial or complete), class or major.

2. Search Students; for a Teacher, restrict to students of assigned sections via Enrollment.

3. Return ID, full name, class, major and academic status, or a no-result notice.

3.2 Filter Students

1. Receive class, academic cohort, major and academic status criteria.

2. Apply the filter with the same scope rule as 3.1.

3. Return the filtered list or a no-result notice.

3.3 View Student Detail

1. Receive the selected student.

2. For a Teacher, check the student belongs to an assigned section.

3. Return personal information, contact data, course sections and academic status.

6.5. Business rules

3.1 Search Students

- Teachers see only students in assigned sections.
- Results are paginated.

3.2 Filter Students

- Same access restriction as 3.1.

3.3 View Student Detail

- A Teacher without permission gets a "no permission" message.
- The record is read-only for Teachers.

7. Level 2 – Process 4.0 Course & Section Management

7.1. Summary

Managing courses, creating sections, viewing offered sections and section student lists.

7.2. Sub-processes and data stores

No.	Sub-process	Description	Data stores

4.1	Manage Courses	Adds, edits and deletes courses.	Courses
4.2	Create Course Sections	Creates a course section for a semester.	Courses; Accounts
4.3	View Offered Sections	Lists the sections offered in a semester.	Courses
4.4	View Section Student Lists	Shows the students registered in a section.	Courses; Enrollment

7.3. Data flows

No.	Data flow	From	To	Description
1	Course Data	Administrator	4.1	Course code, name, credit hours, class periods.
2	Course Result	4.1	Administrator	Success or error message of the course operation.
3	Section Data	Administrator	4.2	Course, semester, schedule, classroom, maximum capacity and teacher.
4	Section Result	4.2	Administrator	Success or error message (e.g. schedule conflict).
5	Section View Request	Administrator, Teacher	4.3	Request to view offered sections.
6	Offered Sections	4.3	Administrator, Teacher	Sections offered in the semester.
7	List Request	Administrator, Teacher	4.4	Request for the student list of a section.
8	Section Student List	4.4	Administrator	Students of a section with current

				enrollment and capacity.
9	Assigned Section Lists	4.4	Teacher	Student lists of the teacher's assigned sections.

7.4. Processing description

4.1 Manage Courses

1. Receive course code, name, credit hours, class periods.

2. Validate the data and uniqueness of the course code.

3. Add/update/delete the course and return the result.

4.2 Create Course Sections

1. Receive course, semester, schedule, classroom, maximum capacity and teacher.

2. Check the teacher (from Accounts) and the schedule for conflicts.

3. Create the section and return the result.

4.3 View Offered Sections

1. Receive the request.

2. Read the offered sections of the semester.

3. Return the list.

4.4 View Section Student Lists

1. Receive the request.

2. Read the section and its enrollments.

3. Return student ID, full name, class, registration date, current enrollment and maximum capacity.

7.5. Business rules

4.1 Manage Courses

- Course code is unique.
- A course with sections or grades cannot be deleted.

4.2 Create Course Sections

- Section code is unique.
- No schedule conflict for the teacher or classroom.

4.3 View Offered Sections

- Available to all roles (Students via process 4.0).

4.4 View Section Student Lists

- Administrator can view any section; Teacher only assigned sections.
- Lists are paginated.

8. Level 2 – Process 5.0 Course Registration

8.1. Summary

Viewing open sections, registering, cancelling and viewing registered sections.

8.2. Sub-processes and data stores

No.	Sub-process	Description	Data stores
5.1	View Open Sections	Shows sections open for registration.	Courses; Enrollment
5.2	Validate & Register	Validates and records a registration.	Courses; Enrollment; Grades
5.3	Cancel Registration	Cancels an existing registration.	Courses; Enrollment
5.4	View Registered Sections	Shows the student's registered sections and schedule.	Enrollment

8.3. Data flows

No.	Data flow	From	To	Description
1	Section View Request	Student	5.1	Request to view offered sections.
2	Open Sections List	5.1	Student	Open sections with credits, teacher, schedule and remaining capacity.
3	Registration Request	Student	5.2	Section the student wants to register for.
4	Registration Result	5.2	Student	Accepted, or rejected with a reason.
5	Cancellation Request	Student	5.3	Registered section the student wants to cancel.
6	Cancellation Result	5.3	Student	
7	Schedule View Request	Student	5.4	Request to view registered sections.
8	Registered Sections	5.4	Student	Registered sections, schedule and registration date.

8.4. Processing description

5.1 View Open Sections

1. Receive the request.

2. Read open sections and compute remaining seats.

3. Return section code, course, credits, teacher, schedule and remaining capacity.

5.2 Validate & Register

1. Receive the registration request.

2. Check registration period, remaining capacity, duplicate registration, schedule conflict and prerequisites (passed courses read from Grades).

3. If valid, create the Enrollment and update the section's remaining capacity.

4. Return the result or the rejection reason.

5.3 Cancel Registration

1. Receive the request and check the registration belongs to the student.

2. Check the cancellation deadline.

3. Mark the Enrollment as cancelled and increase the remaining capacity.

4. Return the result.

5.4 View Registered Sections

1. Receive the request.

2. Read the student's active enrollments.

3. Return section code, course, teacher, schedule and registration date.

8.5. Business rules

5.1 View Open Sections

- If no section is open, a message is shown.

5.2 Validate & Register

- Enrollment must never exceed maximum capacity, even with concurrent requests.
- Invalid requests are rejected with a reason.

5.3 Cancel Registration

- Only before the cancellation deadline.
- A student can cancel only his/her own registrations.

5.4 View Registered Sections

- A student sees only his/her own sections.

9. Level 2 – Process 6.0 Grade Management

9.1. Summary

Entering/editing grades, calculating results and GPA, locking/unlocking gradebooks, viewing grades.

9.2. Sub-processes and data stores

No.	Sub-process	Description	Data stores
6.1	Enter & Edit Grades	Teacher enters or edits component grades.	Enrollment; Grades
6.2	Calculate Grades & GPA	Computes final grade, letter grade, four-point grade and semester GPA.	Courses; Grades
6.3	Lock / Unlock Gradebook	Locks a gradebook (Teacher) or unlocks it (Administrator).	Grades
6.4	View Grades	Student views own results.	Grades

9.3. Data flows

No.	Data flow	From	To	Description
1	Component Grades	Teacher	6.1	Attendance, midterm and final scores.
2	Grade Entry Result	6.1	Teacher	Success, or the list of invalid grades / locked gradebook.
3	Validated Grades	6.1	6.2	Scores that passed range and lock checks, sent for calculation.
4	Calculated results	6.2	Grades	Final grade, letter grade, four-point

				grade and semester GPA.
5	Lock Request	Teacher	6.3	Request to lock the gradebook of a section.
6	Unlock Request	Administrator	6.3	Gradebook to unlock and the reason.
7	Lock Status	6.3	Teacher	Locked, or a message that grades are missing.
8	Unlock Status	6.3	Administrator	Result of the unlock.
9	Grade View Request	Student	6.4	Request to view own grades.
10	Grades & GPA	6.4	Student	Component grades, final grades and semester GPA.

9.4. Processing description

6.1 Enter & Edit Grades

1. Read the section's students from Enrollment.

2. Receive attendance, midterm and final scores.

3. Check the gradebook is not locked and each score is in the valid range.

4. Save the scores and send Validated Grades to 6.2; log edits (editor, time, old/new).

5. Return the entry result.

6.2 Calculate Grades & GPA

1. Apply the weights (attendance, midterm, final; total 100%) to compute the final grade.

2. Convert it to a letter grade (A–F) and a four-point grade.

3. Compute the semester GPA, weighted by credit hours.

4. Store the results.

6.3 Lock / Unlock Gradebook

1. For locking: check all students have grades, set status to Locked and publish grades to students.

2. For unlocking: require a reason, set status to Unlocked and record the history (who, when, why).

3. Return the status.

6.4 View Grades

1. Receive the request.

2. Read the student's grades of locked gradebooks.

3. Return component grades, final grades and semester GPA.

9.5. Business rules

6.1 Enter & Edit Grades

- Only the assigned teacher; only while the gradebook is unlocked.
- Invalid scores are listed and rejected.

6.2 Calculate Grades & GPA

- GPA = Σ(four-point grade × credit hours) / Σ credit hours.

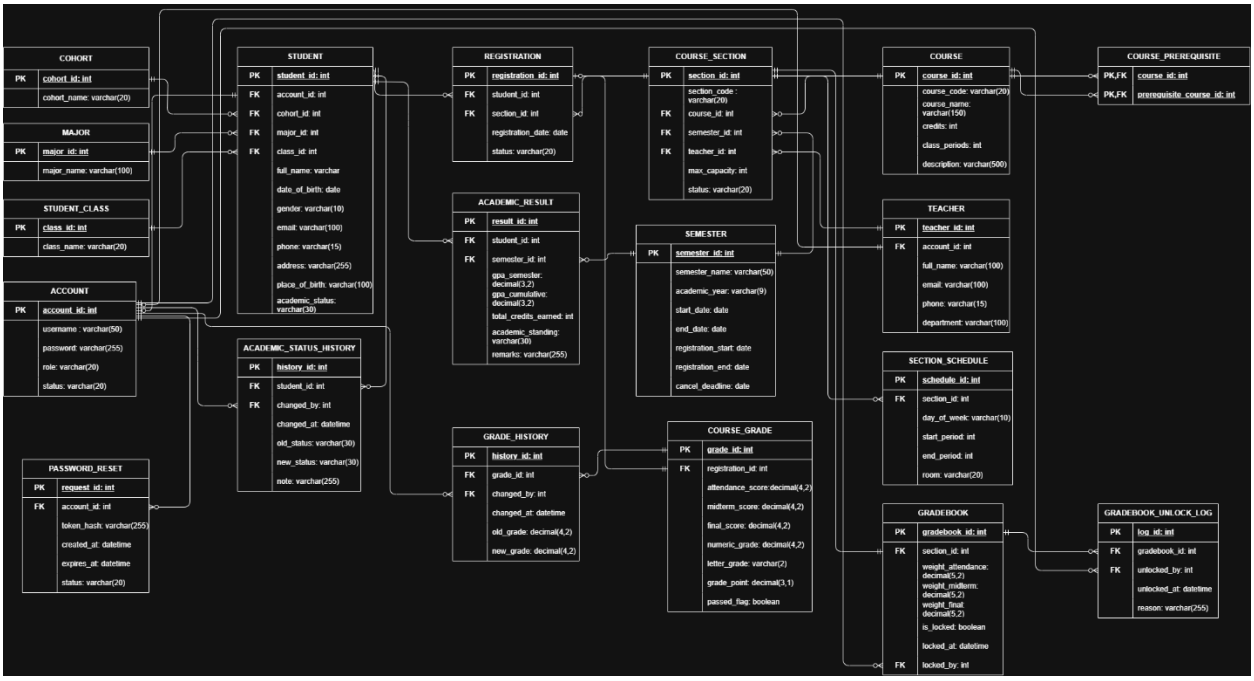
6.3 Lock / Unlock Gradebook

- Teacher locks only own sections; only the Administrator can unlock.

6.4 View Grades

- Only own grades; only grades from locked gradebooks

VI. Data Model



1. Overview

1.1. List of Tables

No.	Table	Description
1	ACCOUNT	Stores the login account of every user of the system (Student, Teacher or Administrator).
2	PASSWORD_RESET	Stores the password-reset requests made by users who forgot their password.
3	COHORT	Lookup table of academic cohorts (intake years).
4	MAJOR	Lookup table of majors.
5	STUDENT_CLASS	Lookup table of student classes.
6	STUDENT	Stores the personal and academic information of students.
7	ACADEMIC_STATUS_HISTORY	Keeps the history of changes of a student's academic status.
8	TEACHER	Stores the information of teachers.

9	COURSE	Stores the courses (subjects) of the curriculum, independent of semester.
10	COURSE_PREREQUISITE	Link table that stores which courses must be completed before another course.
11	SEMESTER	Stores the semesters and their registration and cancellation periods.
12	COURSE_SECTION	Stores the course sections offered for a course in a semester.
13	SECTION_SCHEDULE	Stores the class meetings (day, periods and room) of a course section.
14	REGISTRATION	Stores the registration of a student for a course section.
15	GRADEBOOK	Stores the gradebook of a course section: component weights and lock status.
16	GRADEBOOK_UNLOCK_LOG	Keeps the history of gradebook unlocks.
17	COURSE_GRADE	Stores the grades of one registration: component scores, final grade, letter grade and grade point.
18	GRADE_HISTORY	Keeps the history of changes of a grade.
19	ACADEMIC_RESULT	Stores the semester result of a student: semester GPA, cumulative GPA and academic standing.

1.2. List of Relationships

No.	**Tables**	**Multiplicity**	**Meaning**
1	ACCOUNT – STUDENT	1 – 0..1	One account belongs to at most one student.
2	ACCOUNT – TEACHER	1 – 0..1	One account belongs to at most one teacher.
3	ACCOUNT – PASSWORD_RESET	1 – 0..*	An account can make many reset requests.
4	ACCOUNT – ACADEMIC_STATUS_HISTORY / GRADE_HISTORY /	1 – 0..*	The account that made a change, lock or unlock.

	GRADEBOOK / GRADEBOOK_UNLOCK_LOG		
5	COHORT, MAJOR, STUDENT_CLASS – STUDENT	1 – 0..*	A student has one cohort, major and class.
6	STUDENT – REGISTRATION	1 – 0..*	A student registers for many sections.
7	STUDENT – ACADEMIC_RESULT	1 – 0..*	A student has a result per semester.
8	STUDENT – ACADEMIC_STATUS_HISTORY	1 – 0..*	A student has a history of status changes.
9	COURSE – COURSE_SECTION	1 – 0..*	A course is offered as many sections.
10	COURSE – COURSE_PREREQUISITE	1 – 0..*	Prerequisite relationship of courses.
11	SEMESTER – COURSE_SECTION	1 – 0..*	Sections are held in a semester.
12	SEMESTER – ACADEMIC_RESULT	1 – 0..*	Results belong to a semester.
13	TEACHER – COURSE_SECTION	1 – 0..*	A teacher teaches many sections.
14	COURSE_SECTION – REGISTRATION	1 – 0..*	A section has many registrations.
15	COURSE_SECTION – SECTION_SCHEDULE	1 – 0..*	A section has many class meetings.
16	COURSE_SECTION – GRADEBOOK	1 – 1	A section has one gradebook.
17	REGISTRATION – COURSE_GRADE	1 – 0..1	A registration has at most one grade.
18	COURSE_GRADE – GRADE_HISTORY	1 – 0..*	A grade has a history of changes.
19	GRADEBOOK – GRADEBOOK_UNLOCK_LOG	1 – 0..*	A gradebook has a history of unlocks.

2. Table ACCOUNT

2.1. Summary

Stores the login account of every user of the system (Student, Teacher or Administrator).

2.2. Attributes

Column	Data type	Key	Description
account_id	INT	PK	Unique identifier of the account.
username	VARCHAR(50)		Login name (student ID, teacher ID or administrator ID). Must be unique.
password	VARCHAR(255)		Password, stored in encrypted (hashed) form.
role	VARCHAR(20)		Role of the user: STUDENT, TEACHER or ADMINISTRATOR.
status	VARCHAR(20)		Account status: ACTIVE, LOCKED or INACTIVE.

2.3. Relationships

One account belongs to at most one STUDENT and at most one TEACHER (one-to-one).
One account has many PASSWORD_RESET requests.
One account can be recorded as the editor in ACADEMIC_STATUS_HISTORY and GRADE_HISTORY, as the locker of a GRADEBOOK and as the unlocker in GRADEBOOK_UNLOCK_LOG.

2.4. Special requirements

Passwords must be stored in encrypted form and an initial password is randomly generated.
Accounts with status LOCKED or INACTIVE cannot log in.

3. Table PASSWORD_RESET

3.1. Summary

Stores the password-reset requests made by users who forgot their password.

3.2. Attributes

Column	Data type	Key	Description
request_id	INT	PK	Unique identifier of the request.
account_id	INT	FK	Account that requested the reset (ACCOUNT).
token_hash	VARCHAR(255)		Hash of the reset token sent to the user's email.
created_at	DATETIME		Time the request was created.
expires_at	DATETIME		Time after which the token is no longer valid.
status	VARCHAR(20)		PENDING, COMPLETED or EXPIRED.

3.3. Relationships

Many PASSWORD_RESET requests belong to one ACCOUNT.

3.4. Special requirements

The token itself is not stored, only its hash.
An expired or completed request cannot be used again.
expires_at must be later than created_at.

4. Table COHORT

4.1. Summary

Lookup table of academic cohorts (intake years).

4.2. Attributes

Column	Data type	Key	Description
cohort_id	INT	PK	Unique identifier of the cohort.
cohort_name	VARCHAR(20)		Name of the cohort, for example 2024.

4.3. Relationships

One cohort has many STUDENT records.

4.4. Special requirements

None

5. Table MAJOR

5.1. Summary

Lookup table of majors.

5.2. Attributes

Column	Data type	Key	Description
major_id	INT	PK	Unique identifier of the major.
major_name	VARCHAR(100)		Name of the major, for example Information Technology.

5.3. Relationships

One major has many STUDENT records.

5.4. Special requirements

None

6. Table STUDENT_CLASS

6.1. Summary

Lookup table of student classes.

6.2. Attributes

Column	Data type	Key	Description
class_id	INT	PK	Unique identifier of the class.
class_name	VARCHAR(20)		Name of the class, for example IT24A.

6.3. Relationships

One class has many STUDENT records.

6.4. Special requirements

None

7. Table STUDENT

7.1. Summary

Stores the personal and academic information of students.

7.2. Attributes

Column	Data type	Key	Description
student_id	INT	PK	Unique identifier of the student.
account_id	INT	FK	Login account of the student (ACCOUNT).
cohort_id	INT	FK	Academic cohort (COHORT).
major_id	INT	FK	Major (MAJOR).
class_id	INT	FK	Student class (STUDENT_CLASS).
full_name	VARCHAR(100)		Full name.

date_of_birth	DATE		Date of birth.
gender	VARCHAR(10)		MALE, FEMALE or OTHER.
email	VARCHAR(100)		Email address.
phone	VARCHAR(15)		Phone number.
address	VARCHAR(255)		Contact address.
place_of_birth	VARCHAR(100)		Place of birth.
academic_status	VARCHAR(30)		ENROLLED, WITHDRAWN, TEMPORARILY_SUSPENDED or GRADUATED.

7.3. Relationships

One STUDENT belongs to one ACCOUNT, one COHORT, one MAJOR and one STUDENT_CLASS.
One STUDENT has many REGISTRATION, ACADEMIC_RESULT and ACADEMIC_STATUS_HISTORY records.

7.4. Special requirements

The student ID is unique and cannot be changed after creation.
A student can update only email, phone and address; all other columns are changed by the Administrator.
Every change of academic_status is logged in ACADEMIC_STATUS_HISTORY.

8. Table ACADEMIC_STATUS_HISTORY

8.1. Summary

Keeps the history of changes of a student's academic status.

8.2. Attributes

Column	Data type	Key	Description
history_id	INT	PK	Unique identifier of the history record.
student_id	INT	FK	Student whose status changed (STUDENT).
changed_by	INT	FK	Account of the administrator who made the change (ACCOUNT).
changed_at	DATETIME		Time of the change.
old_status	VARCHAR(30)		Status before the change.
new_status	VARCHAR(30)		Status after the change.
note	VARCHAR(255)		Optional note of the administrator.

8.3. Relationships

Many records belong to one STUDENT.
Many records refer to one ACCOUNT (the editor).

8.4. Special requirements

Records are only added, never edited or deleted.
old_status and new_status must be different valid academic statuses.

9. Table TEACHER

9.1. Summary

Stores the information of teachers.

9.2. Attributes

Column	Data type	Key	Description
teacher_id	INT	PK	Unique identifier of the teacher.

account_id	INT	FK	Login account of the teacher (ACCOUNT).
full_name	VARCHAR(100)		Full name.
email	VARCHAR(100)		Email address (can be used to log in).
phone	VARCHAR(15)		Phone number.
department	VARCHAR(100)		Department the teacher belongs to.

9.3. Relationships

One TEACHER belongs to one ACCOUNT.
One TEACHER is assigned to many COURSE_SECTION records.

9.4. Special requirements

A teacher can update only phone and email; other data is changed by the Administrator.

10. Table COURSE

10.1. Summary

Stores the courses (subjects) of the curriculum, independent of semester.

10.2. Attributes

Column	Data type	Key	Description
course_id	INT	PK	Unique identifier of the course.
course_code	VARCHAR(20)		Course code, for example INT2203.
course_name	VARCHAR(150)		Course name.
credits	INT		Credit hours.
class_periods	INT		Number of class periods.

description	VARCHAR(500)		Optional description.

10.3. Relationships

One COURSE has many COURSE_SECTION records.
A COURSE can have many prerequisites and be a prerequisite of many courses (through COURSE_PREREQUISITE).

10.4. Special requirements

course_code is unique.
A course that has course sections or grades cannot be deleted.

11. Table COURSE_PREREQUISITE

11.1. Summary

Link table that stores which courses must be completed before another course.

11.2. Attributes

Column	Data type	Key	Description
course_id	INT	PK, FK	The course that has a prerequisite (COURSE).
prerequisite_course_id	INT	PK, FK	The required course (COURSE).

11.3. Relationships

Both columns refer to COURSE (many-to-many relationship of COURSE with itself).

11.4. Special requirements

The pair (course_id, prerequisite_course_id) is unique.
A course cannot be its own prerequisite, directly or indirectly.

12. Table SEMESTER

12.1. Summary

Stores the semesters and their registration and cancellation periods.

12.2. Attributes

Column	Data type	Key	Description
semester_id	INT	PK	Unique identifier of the semester.
semester_name	VARCHAR(50)		Name, for example Semester 1.
academic_year	VARCHAR(9)		Academic year, for example 2026-2027.
start_date	DATE		First day of the semester.
end_date	DATE		Last day of the semester.
registration_start	DATE		First day of the registration period.
registration_end	DATE		Last day of the registration period.
cancel_deadline	DATE		Last day a registration can be cancelled.

12.3. Relationships

One SEMESTER has many COURSE_SECTION and ACADEMIC_RESULT records.

12.4. Special requirements

start_date must be before end_date and registration_start before registration_end. Registration is accepted only within the registration period and cancellation only until cancel_deadline.

13. Table COURSE_SECTION

13.1. Summary

Stores the course sections offered for a course in a semester.

13.2. Attributes

Column	Data type	Key	Description
section_id	INT	PK	Unique identifier of the section.
section_code	VARCHAR(20)		Section code, for example INT2203-01.
course_id	INT	FK	Course (COURSE).
semester_id	INT	FK	Semester (SEMESTER).
teacher_id	INT	FK	Responsible teacher (TEACHER).
max_capacity	INT		Maximum number of students.
status	VARCHAR(20)		Status of the section, for example OPEN or CLOSED.

13.3. Relationships

Many COURSE_SECTION records belong to one COURSE, one SEMESTER and one TEACHER.
One COURSE_SECTION has many REGISTRATION and SECTION_SCHEDULE records and exactly one GRADEBOOK.

13.4. Special requirements

section_code is unique.
The number of active registrations never exceeds max_capacity.
The teacher and the classroom must have no schedule conflict.

14. Table SECTION_SCHEDULE

14.1. Summary

Stores the class meetings (day, periods and room) of a course section.

14.2. Attributes

Column	Data type	Key	Description
schedule_id	INT	PK	Unique identifier of the schedule record.

section_id	INT	FK	Course section (COURSE_SECTION).
day_of_week	VARCHAR(10)		Day of the week.
start_period	INT		First period.
end_period	INT		Last period.
room	VARCHAR(20)		Classroom, for example B2-301.

14.3. Relationships

Many SECTION_SCHEDULE records belong to one COURSE_SECTION.

14.4. Special requirements

start_period must not be greater than end_period.
The same teacher or room cannot have overlapping schedules; a student cannot register two sections with overlapping schedules.

14.5. Related use-cases

Create course sections, Register for course sections.

15. Table REGISTRATION

15.1. Summary

Stores the registration of a student for a course section.

15.2. Attributes

Column	Data type	Key	Description
registration_id	INT	PK	Unique identifier of the registration.
student_id	INT	FK	Student (STUDENT).
section_id	INT	FK	Course section (COURSE_SECTION).
registration_date	DATE		Date of registration.

status	VARCHAR(20)		REGISTERED or CANCELLED.

15.3. Relationships

Many REGISTRATION records belong to one STUDENT and one COURSE_SECTION.
One REGISTRATION has at most one COURSE_GRADE.

15.4. Special requirements

A student cannot have two REGISTERED records for the same section.
A registration is changed to CANCELLED only before the cancellation deadline.

16. Table GRADEBOOK

16.1. Summary

Stores the gradebook of a course section: component weights and lock status.

16.2. Attributes

Column	Data type	Key	Description
gradebook_id	INT	PK	Unique identifier of the gradebook.
section_id	INT	FK	Course section (COURSE_SECTION).
weight_attendance	DECIMAL(5, 2)		Weight of attendance, in percent.
weight_midterm	DECIMAL(5, 2)		Weight of the midterm exam, in percent.
weight_final	DECIMAL(5, 2)		Weight of the final exam, in percent.
is_locked	BOOLEAN		True when the gradebook is locked.
locked_at	DATETIME		Time the gradebook was locked.
locked_by	INT	FK	Account of the teacher who locked it (ACCOUNT).

16.3. Relationships

One GRADEBOOK belongs to one COURSE_SECTION (one-to-one).
One GRADEBOOK has many GRADEBOOK_UNLOCK_LOG records.

16.4. Special requirements

The three weights add up to 100.
While is_locked is true, grades of the section cannot be entered or edited, and only the Administrator can unlock it.
Locking is allowed only when all students of the section have grades.

17. Table GRADEBOOK_UNLOCK_LOG

17.1. Summary

Keeps the history of gradebook unlocks.

17.2. Attributes

Column	Data type	Key	Description
log_id	INT	PK	Unique identifier of the log record.
gradebook_id	INT	FK	Unlocked gradebook (GRADEBOOK).
unlocked_by	INT	FK	Account of the administrator (ACCOUNT).
unlocked_at	DATETIME		Time of the unlock.
reason	VARCHAR(255)		Reason for unlocking.

17.3. Relationships

Many records belong to one GRADEBOOK and refer to one ACCOUNT.

17.4. Special requirements

reason is required.
Records are only added, never edited or deleted.

18. Table COURSE_GRADE

18.1. Summary

Stores the grades of one registration: component scores, final grade, letter grade and grade point.

18.2. Attributes

Column	Data type	Key	Description
grade_id	INT	PK	Unique identifier of the grade.
registration_id	INT	FK	Registration being graded (REGISTRATION).
attendance_score	DECIMAL(4, 2)		Attendance score.
midterm_score	DECIMAL(4, 2)		Midterm score.
final_score	DECIMAL(4, 2)		Final examination score.
numeric_grade	DECIMAL(4, 2)		Final course grade, calculated from the weights.
letter_grade	VARCHAR(2)		Letter grade: A, B, C, D or F.
grade_point	DECIMAL(3, 1)		Grade on the four-point scale.
passed_flag	BOOLEAN		True if the student passed the course.

18.3. Relationships

One COURSE_GRADE belongs to one REGISTRATION (at most one grade per registration).
One COURSE_GRADE has many GRADE_HISTORY records.

18.4. Special requirements

Scores must be numbers within the valid range 0–10.

numeric_grade, letter_grade, grade_point and passed_flag are calculated automatically.
Students see the grade only after the gradebook is locked.

19. Table GRADE_HISTORY

19.1. Summary

Keeps the history of changes of a grade.

19.2. Attributes

Column	Data type	Key	Description
history_id	INT	PK	Unique identifier of the history record.
grade_id	INT	FK	Changed grade (COURSE_GRADE).
changed_by	INT	FK	Account of the teacher who made the change (ACCOUNT).
changed_at	DATETIME		Time of the change.
old_grade	DECIMAL(4, 2)		Grade before the change.
new_grade	DECIMAL(4, 2)		Grade after the change.

19.3. Relationships

Many records belong to one COURSE_GRADE and refer to one ACCOUNT.

19.4. Special requirements

Every grade change is logged (editor, time, old grade, new grade).
Records are only added, never edited or deleted.

20. Table ACADEMIC_RESULT

20.1. Summary

Stores the semester result of a student: semester GPA, cumulative GPA and academic standing.

20.2. Attributes

Column	Data type	Key	Description
result_id	INT	PK	Unique identifier of the result.
student_id	INT	FK	Student (STUDENT).
semester_id	INT	FK	Semester (SEMESTER).
gpa_semester	DECIMAL(3, 2)		GPA of the semester.
gpa_cumulative	DECIMAL(3, 2)		Cumulative GPA up to the semester.
total_credits_earned	INT		Credits earned.
academic_standing	VARCHAR(30)		Academic standing.
remarks	VARCHAR(255)		Remarks.

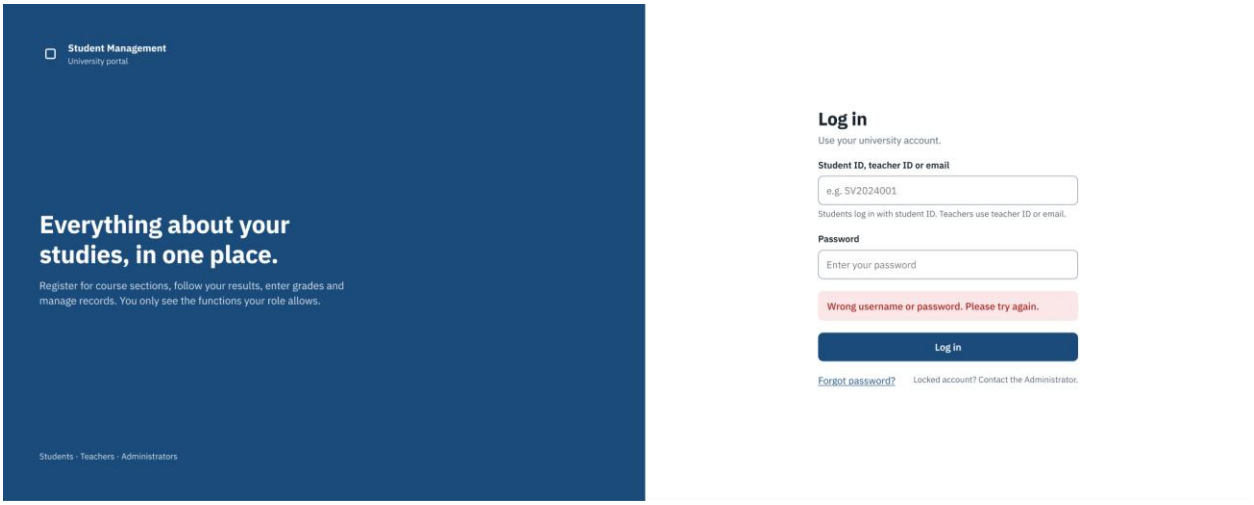
20.3. Relationships

Many ACADEMIC_RESULT records belong to one STUDENT and one SEMESTER.

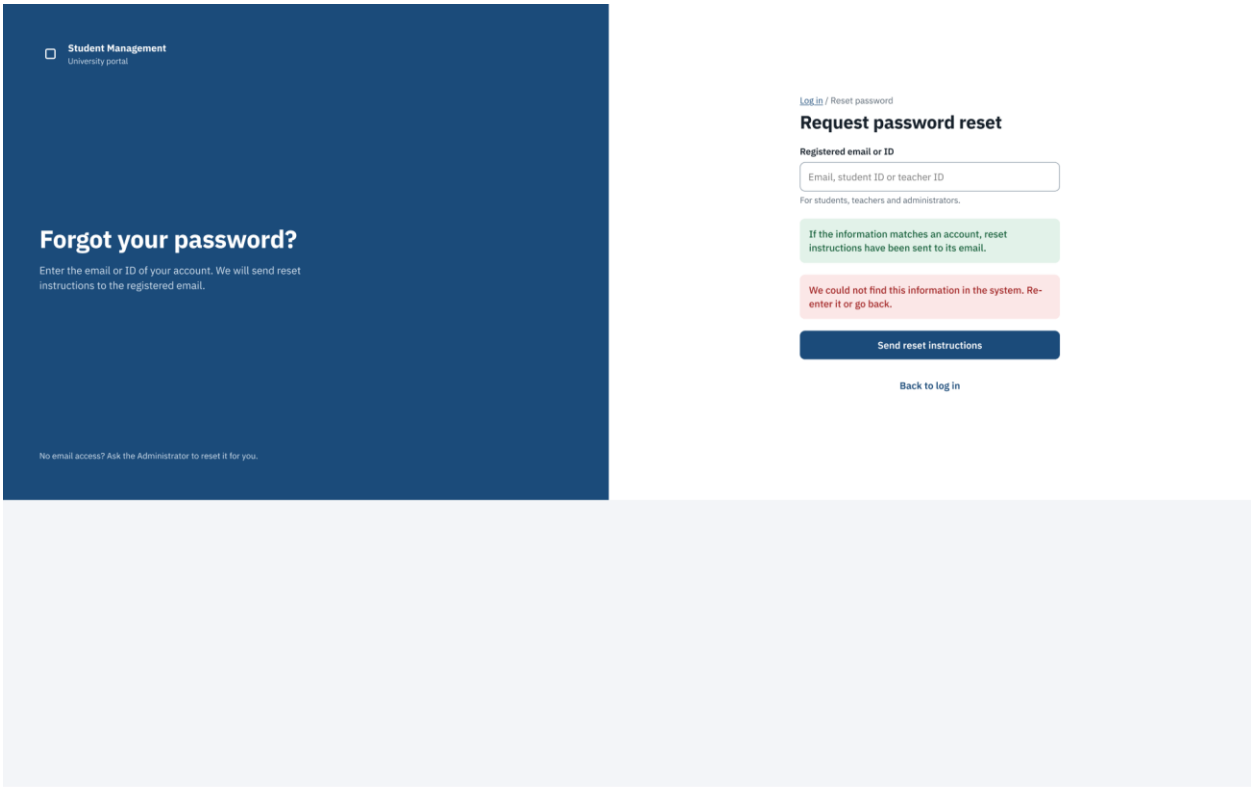
20.4. Special requirements

There is at most one result per student per semester.
gpa_semester is the average of the four-point grades weighted by credit hours.
Only grades of locked gradebooks are counted.

VII. GUI Design



Hình 1: Account Login



Hình 2: Account Reset Password

1. Overview

This document describes the screens of the Student Management System as designed in the GUI mock-ups. After login, every screen uses a left sidebar with the menu of the user's role (and the user's name, ID and a Log out button) and a main content area. Long lists are paginated and statuses are shown as colour badges.

Menu by role:

Student: Personal information, Open course sections, Registered sections, Academic results.

Teacher: Search students, Assigned student lists, Gradebook, Contact details.

Administrator: Accounts, Student records, Search students, Courses, Course sections, Gradebooks.

Screens that appear in these menus but are not in the provided design (Teacher: Assigned student lists, Gradebook, Contact details; Administrator: Accounts, Student records, Search students, Courses, create course section form) are not described here.

2. Account functions (all users)

Functions shared by Students, Teachers and the Administrator: logging in with the university account and requesting a password reset when the password is forgotten. These screens are available before login and do not show the sidebar.

2.1. Log in

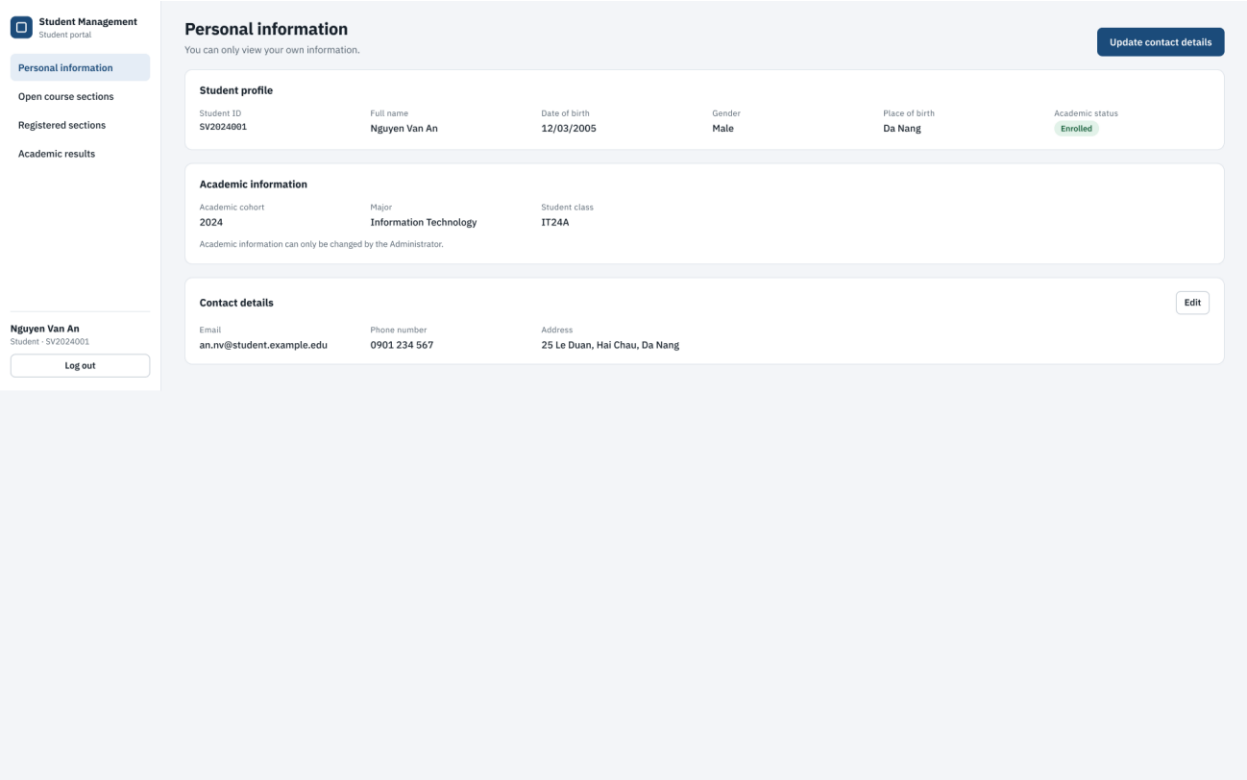
Entry screen of the system. The user enters the account information to log in; the left panel shows the system name and a short welcome message. (Use-case: Log in.)
Student ID, teacher ID or email: Login identifier. Hint: students log in with student ID, teachers with teacher ID or email. Placeholder: e.g. SV2024001. Required.
Password: Password of the account. Required; characters are masked.
Log in: Submits the login information.
Forgot password?: Opens the Request password reset screen.
Locked account note: "Locked account? Contact the Administrator."
Log in → the system checks the information; on success the user goes to the home screen of the role (Student, Teacher or Administrator). Forgot password? → Request password reset screen. Wrong username or password: "Wrong username or password. Please try again." (red message above the button). Locked or inactive account: the system reports that the account cannot be used.

2.2. Request password reset

Lets a user who forgot the password request reset instructions by email. (Use-case: Request password reset.)
Breadcrumb: Log in / Reset password; returns to the login screen.
Registered email or ID: Email, student ID or teacher ID of the account. Required.
Send reset instructions: Submits the request.
Back to log in: Returns to the login screen.
Note: "No email access? Ask the Administrator to reset it for you."
Send reset instructions → the system checks the information and sends reset instructions to the registered email. Back to log in → Log in screen. Success (green): "If the information matches an account, reset instructions have been sent to its email." Error (red): "We could not find this information in the system. Re-enter it or go back."

3. Student functions

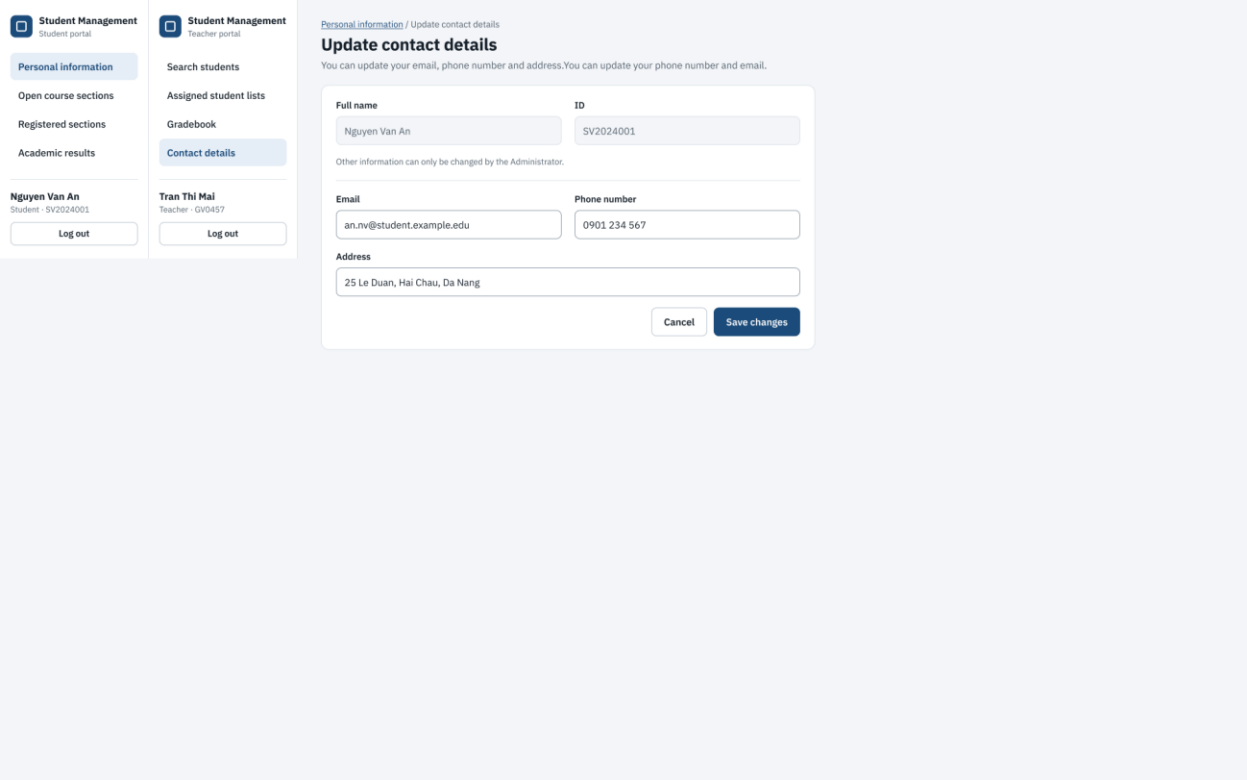
3.1. Student – Personal information



Hình 3: Student View Personal Information

- Shows the student's own information and gives access to updating contact details.
- Sidebar menu: Personal information, Open course sections, Registered sections, Academic results.
- Student profile: Student ID, full name, date of birth, gender, place of birth, academic status
- Academic information: Academic cohort, major, student class. Note: can only be changed by the Administrator.
- Contact details: Email, phone number, address, with an Edit button.
- Update contact details / Edit: Opens the Update contact details screen.
- User box and Log out: Shows name, role and ID of the logged-in user; Log out ends the session.

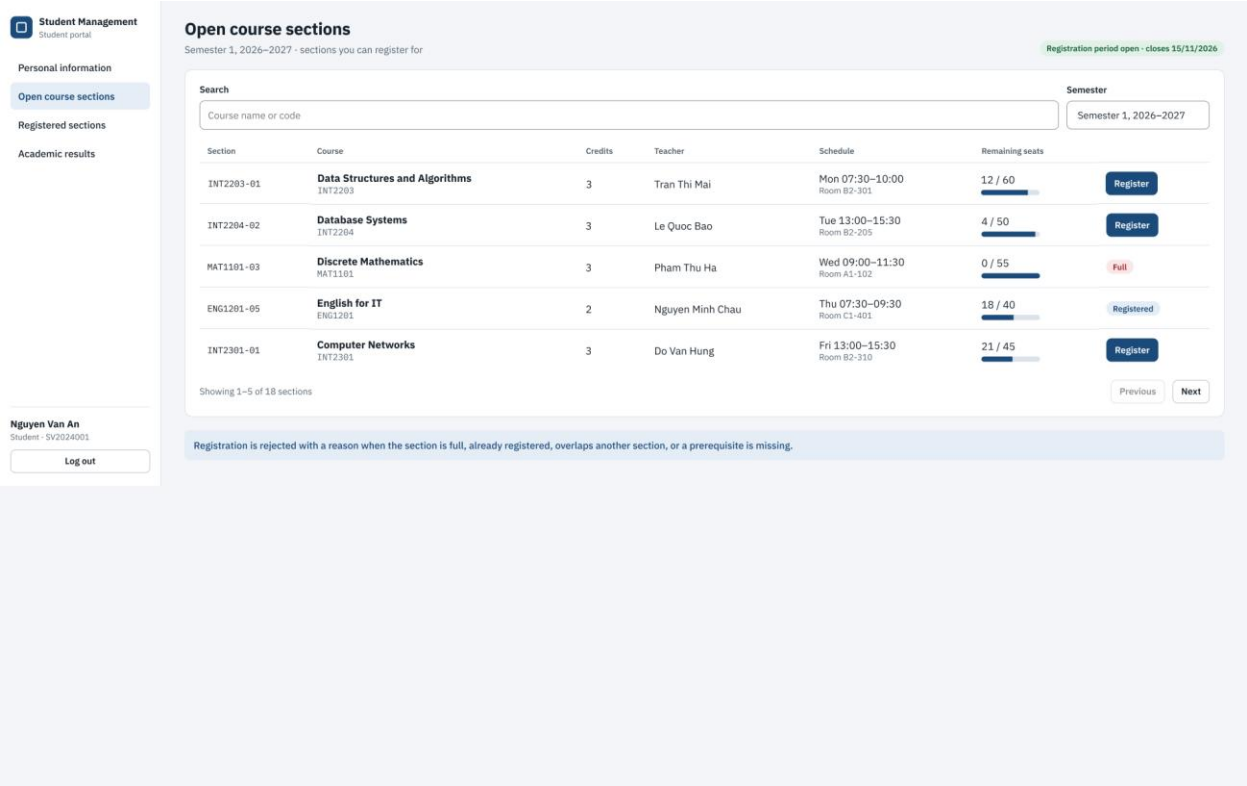
3.2. Student – Update contact details



Hình 4: Student Update Contract Details

On this screen the full name and ID are displayed but disabled, with a note that other information can only be changed by the Administrator. The student can edit the email, phone number and address. Pressing Save changes validates the information, saves it and reports the result; Cancel discards the changes and returns to Personal information. If a field is missing or invalid, an error message is shown and the existing data is kept.

3.3. Open course sections

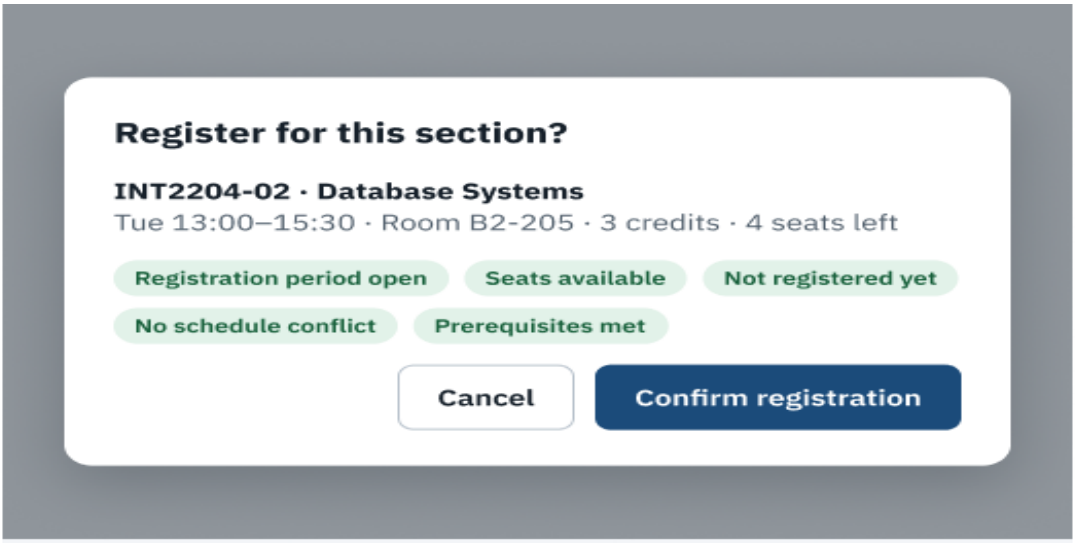


Hình 5: Student Open Course Section

This screen lists the course sections open for registration. A label at the top right shows that the registration period is open and when it closes. The student can search by course name or code and choose a semester. The table shows, for each section, the section code, course name and code, credits, teacher, schedule and room, and the remaining seats out of the maximum with a progress bar. Each row has a Register button; a section with no seats left shows Full instead, and a section the student has already registered shows Registered. The list is paginated, and if no section is open a message is displayed.

Pressing Register opens a confirmation dialog. A note at the bottom explains that a registration is rejected with a reason when the section is full, already registered, overlaps another section or a prerequisite is missing.

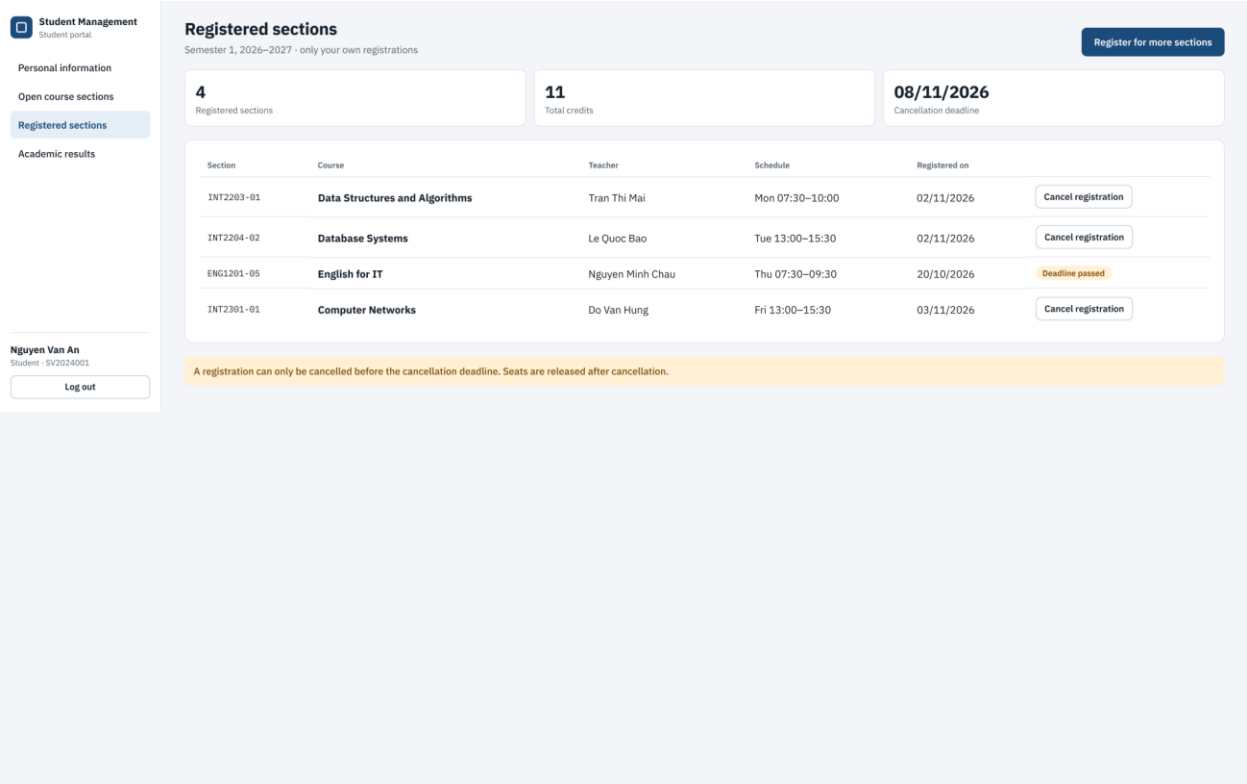
3.4. Comfirm registration dialog



Hình 6: Comfirm registration

This pop-up shows the section code, course name, schedule, room, credits and number of seats left, together with labels for the checks made by the system: registration period open, seats available, not registered yet, no schedule conflict and prerequisites met. The student presses Confirm registration to register or Cancel to close the dialog. After confirmation the system records the registration, reduces the remaining seats and reports the result. If a condition is not met, the registration is rejected and the reason is shown.

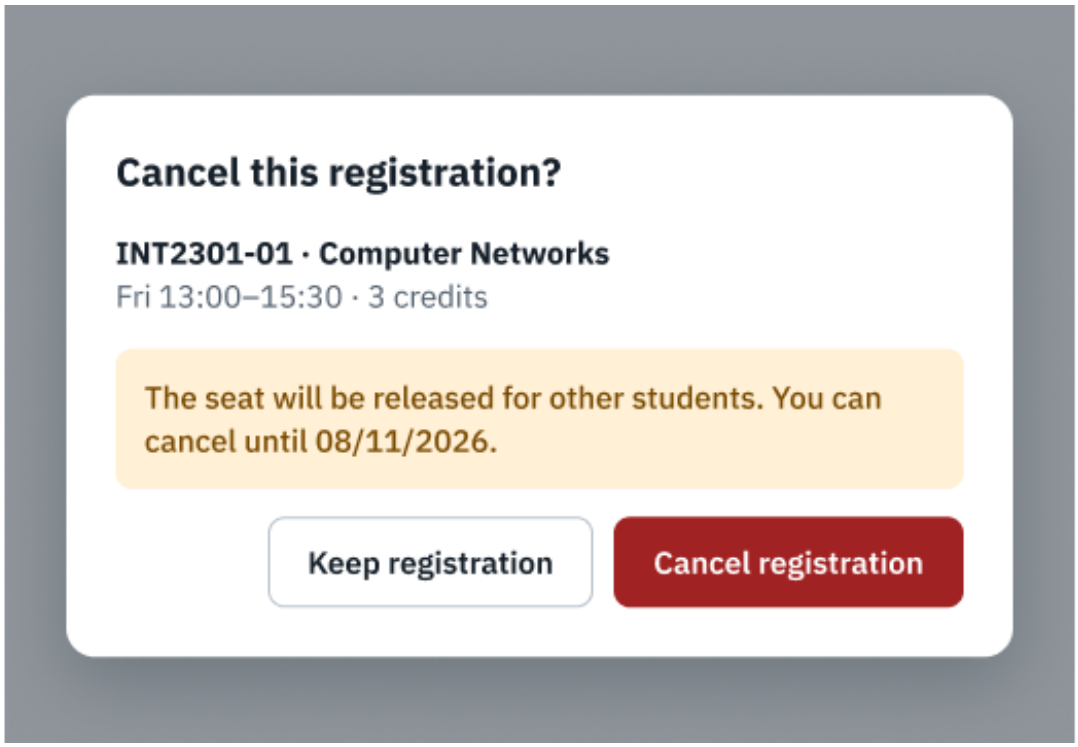
3.5. Registered Section



Hình 7: Student View Registration

This screen shows the sections the student has registered for in the semester, with the number of registered sections, the total credits and the cancellation deadline. For each section it shows the section code, course, teacher, schedule and registration date. Each row has a Cancel registration button, which is replaced by the label Deadline passed when cancellation is no longer allowed. A button leads back to the open sections to register for more. A note reminds the student that a registration can only be cancelled before the deadline and that the seat is released after cancellation. If the student has no registration, a message is shown.

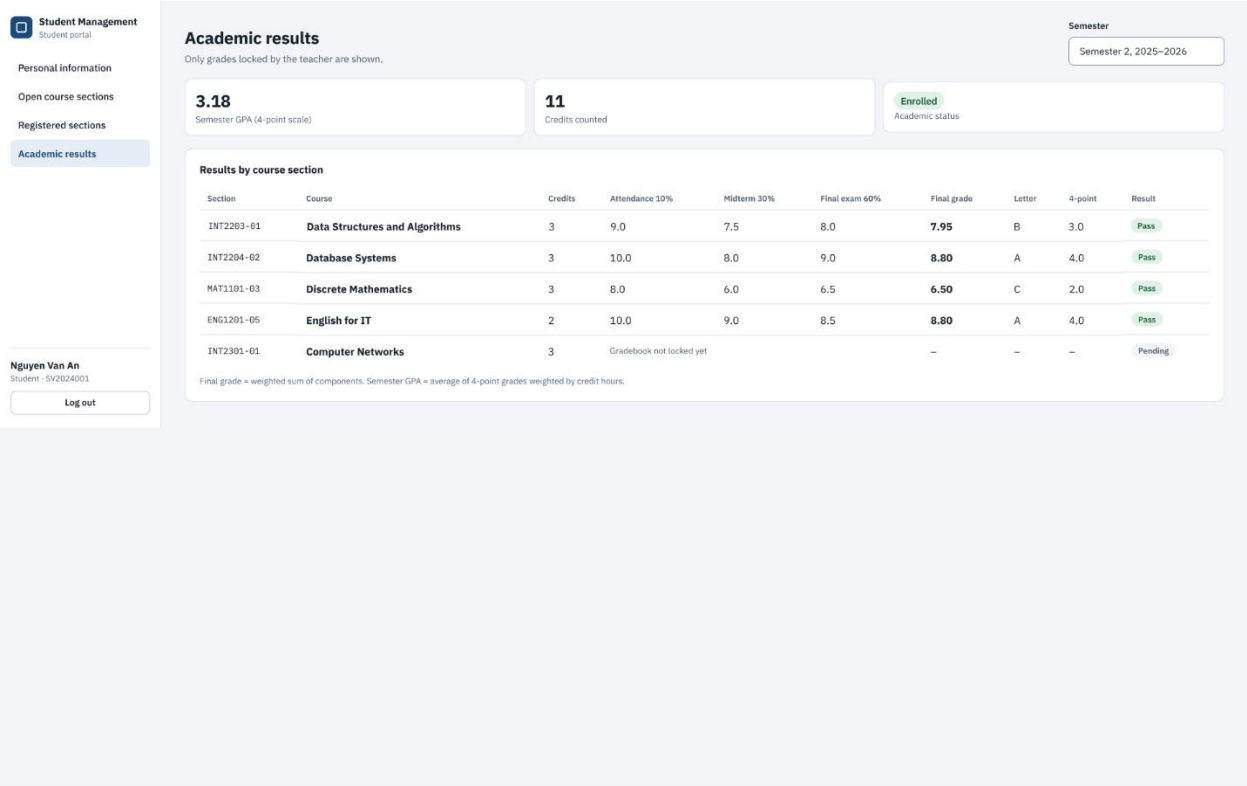
3.6. Cancel Registration dialog



Hình 8: Student Cancel Registration

After the student presses Cancel registration, a dialog shows the section code, course name, schedule and credits, and warns that the seat will be released for other students and gives the last day to cancel. The student confirms with Cancel registration or keeps the registration with Keep registration. After confirmation the system checks the deadline, cancels the registration, increases the remaining seats and reports the result. If the deadline has passed, an error message is shown and the registration is kept.

3.7. Academic result

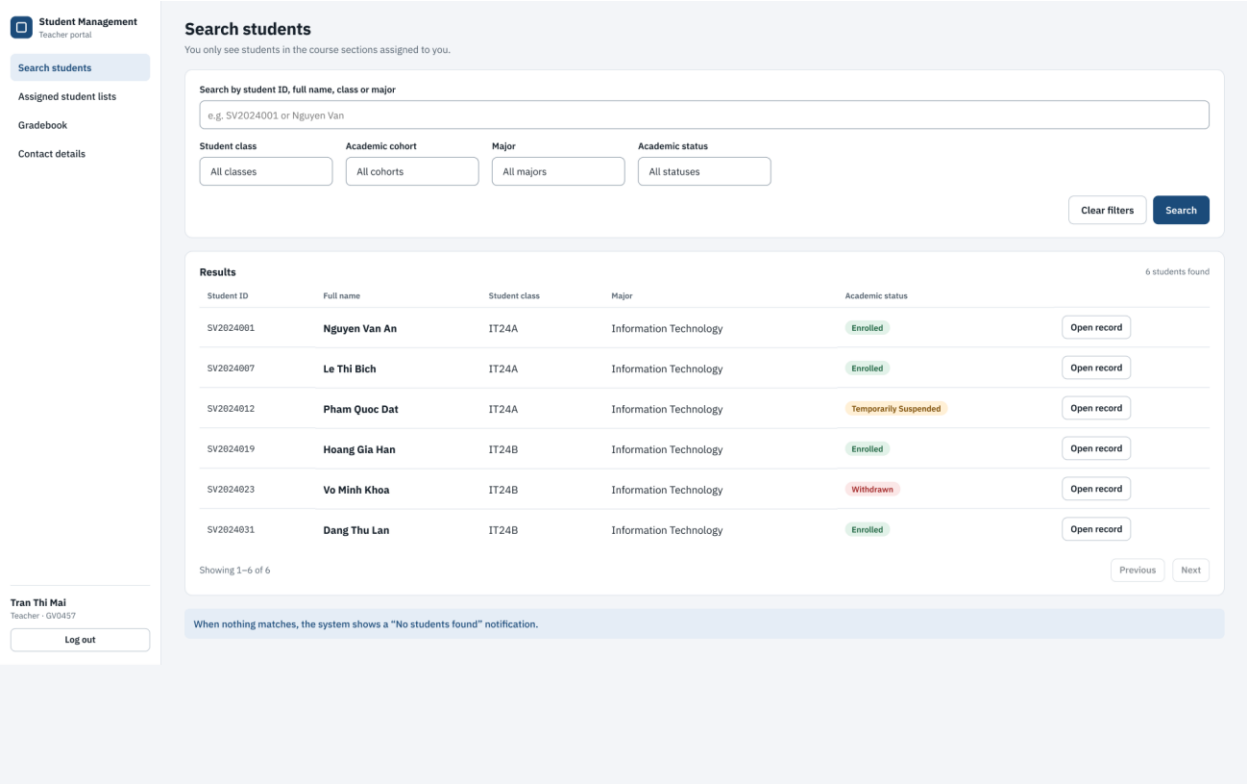


Hình 9: Student View Academic Result

This screen shows the student's results for the selected semester. It displays the semester GPA on the 4-point scale, the credits counted and the academic status. The results table shows, for each section, the section code, course, credits, attendance, midterm and final exam scores with their weights (10%, 30% and 60%), the final grade, the letter grade, the 4-point grade and the result (Pass or Pending). Only grades locked by the teacher are shown; for a section whose gradebook is not locked yet, the table shows "Gradebook not locked yet" and the result is Pending. A note explains that the final grade is the weighted sum of the components and the semester GPA is the average of the 4-point grades weighted by credit hours. If there are no results, a message is shown.

4. Teacher functions

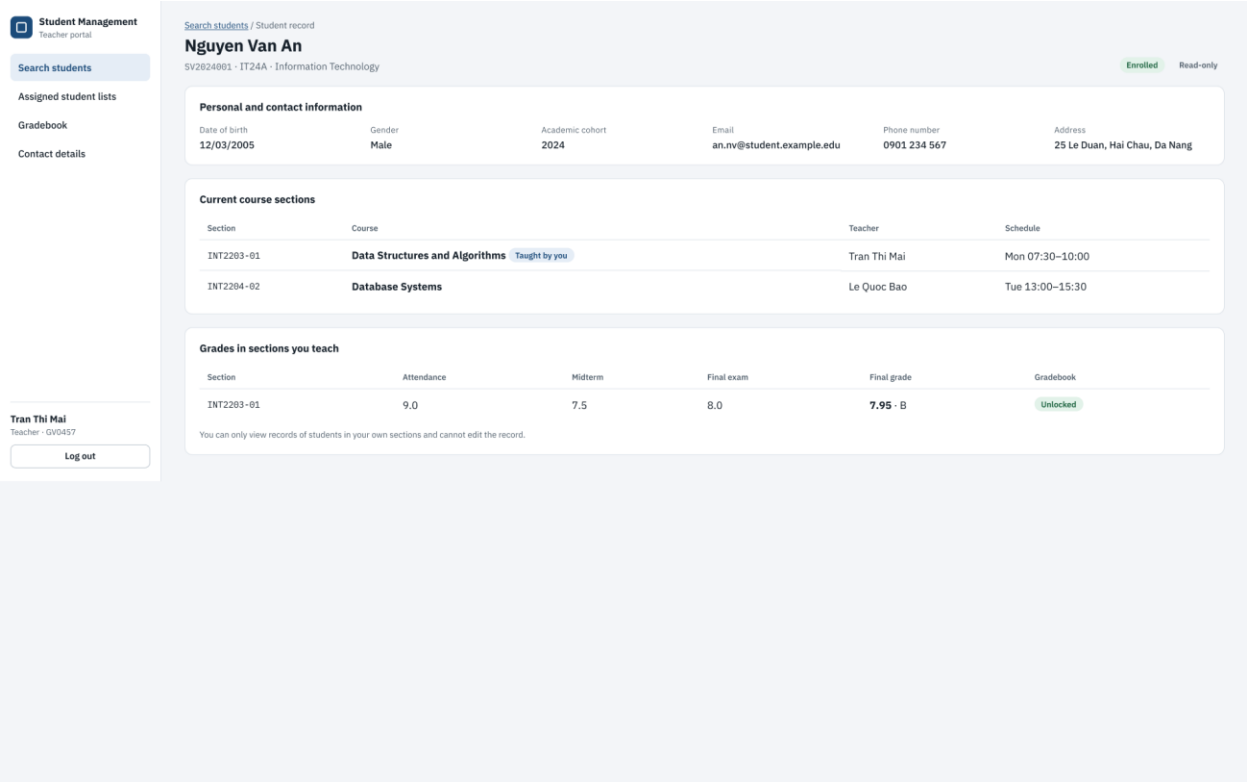
4.1. Search students and filter students



Hình 10: Search and filter student

The teacher can search by student ID, full name, class or major, and filter by student class, academic cohort, major and academic status. Search runs the search and Clear filters resets the criteria. The results show the student ID, full name, student class, major and academic status, with a count of the students found and pagination. Open record opens the student's detailed record. If nothing matches, a "No students found" notification is shown.

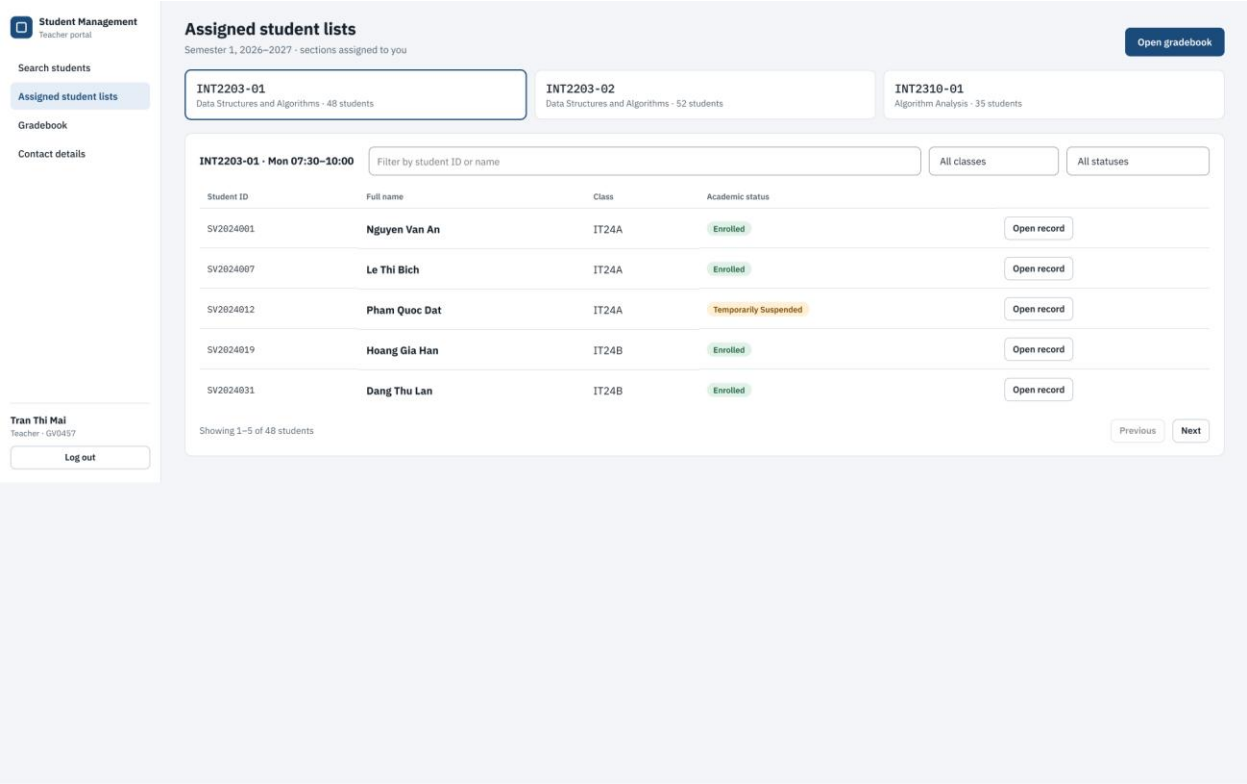
4.2. Student record



Hình 11: Student record

This read-only screen, opened from the search results or an assigned student list, shows the student's name, ID, class, major and academic status. It contains the personal and contact information (date of birth, gender, academic cohort, email, phone number and address), the student's current course sections with a tag on the sections taught by the logged-in teacher, and the grades in the sections the teacher teaches with the gradebook status. The teacher cannot edit the record and can only view students of his or her own sections; otherwise the system reports that the teacher has no permission.

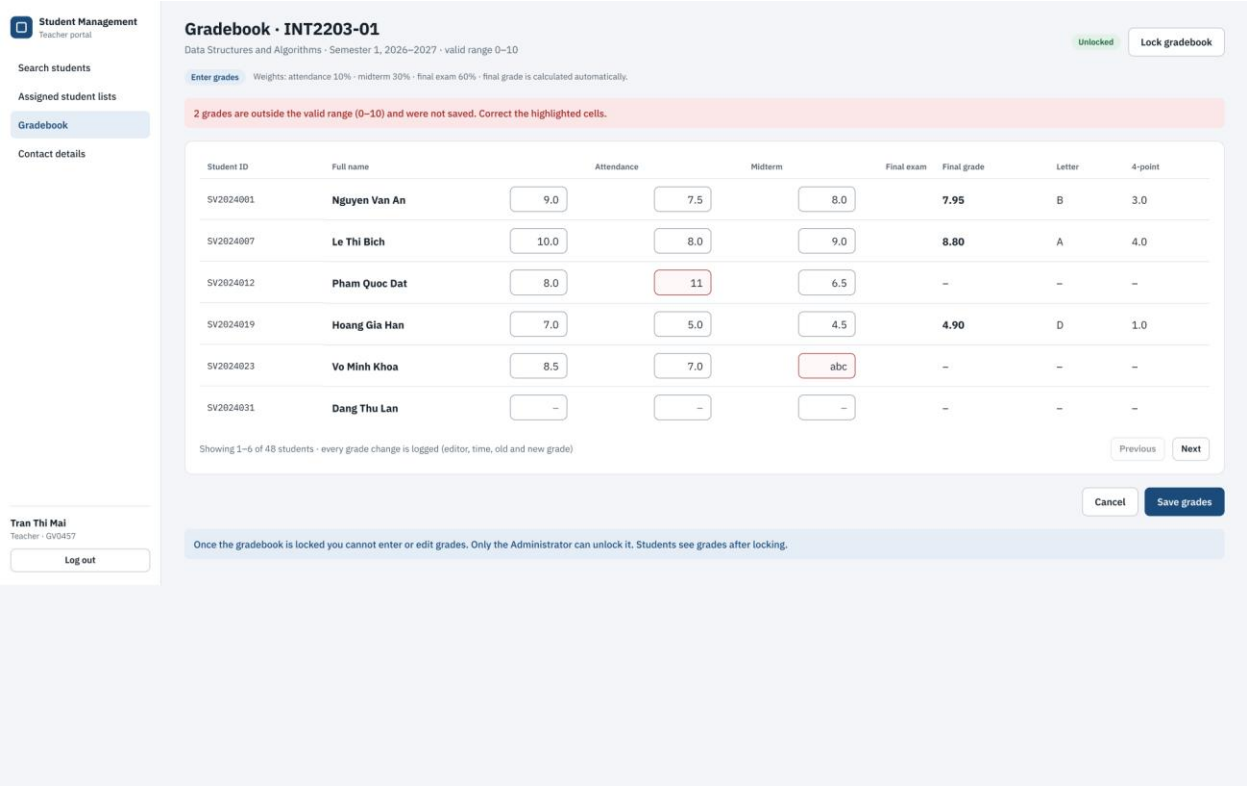
4.3. Assigned student lists



Hình 12: View assigned student

This screen shows the course sections assigned to the teacher in the semester, each with its section code, course name and number of students. After the teacher chooses a section, the list of its students is shown with the student ID, full name, class and academic status, and can be filtered by student ID or name, class and academic status. Open record opens the student's record and Open gradebook leads to the gradebook. The list is paginated. If no section is assigned, a message is shown.

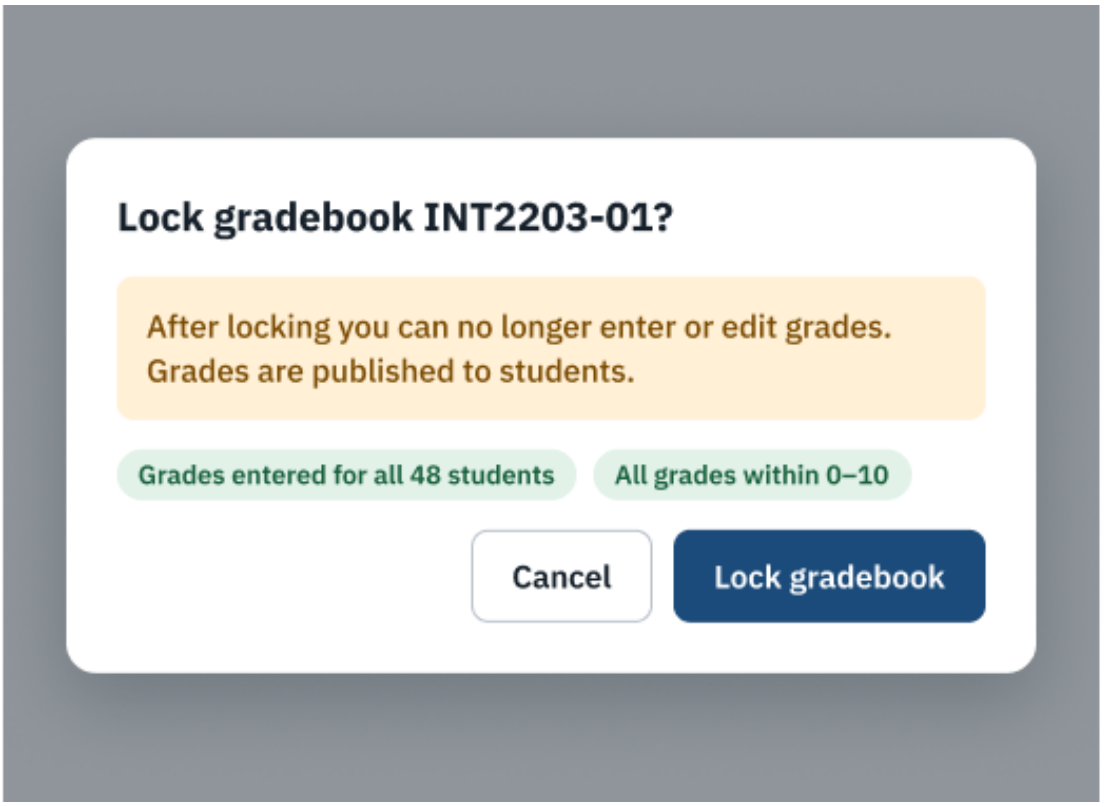
4.4. Edit gradebook



Hình 13: Edit and lock gradebook

This screen shows the gradebook of a section with its status (Unlocked or Locked), the valid grade range 0–10 and the weights of the components (attendance 10%, midterm 30%, final exam 60%). The teacher enters or edits the attendance, midterm and final exam scores of each student. The final grade, letter grade and 4-point grade are calculated automatically and are shown as a dash while the scores are incomplete. The teacher saves with Save grades or discards the changes with Cancel. Before saving, the system checks that every score is a number within the valid range; invalid cells are highlighted and an error message tells the teacher to correct them, and those grades are not saved. Every grade change is logged with the editor, time, old grade and new grade. The Lock gradebook button locks the gradebook. Once it is locked, grades can no longer be entered or edited and only the Administrator can unlock it.

4.5. Lock gradebook

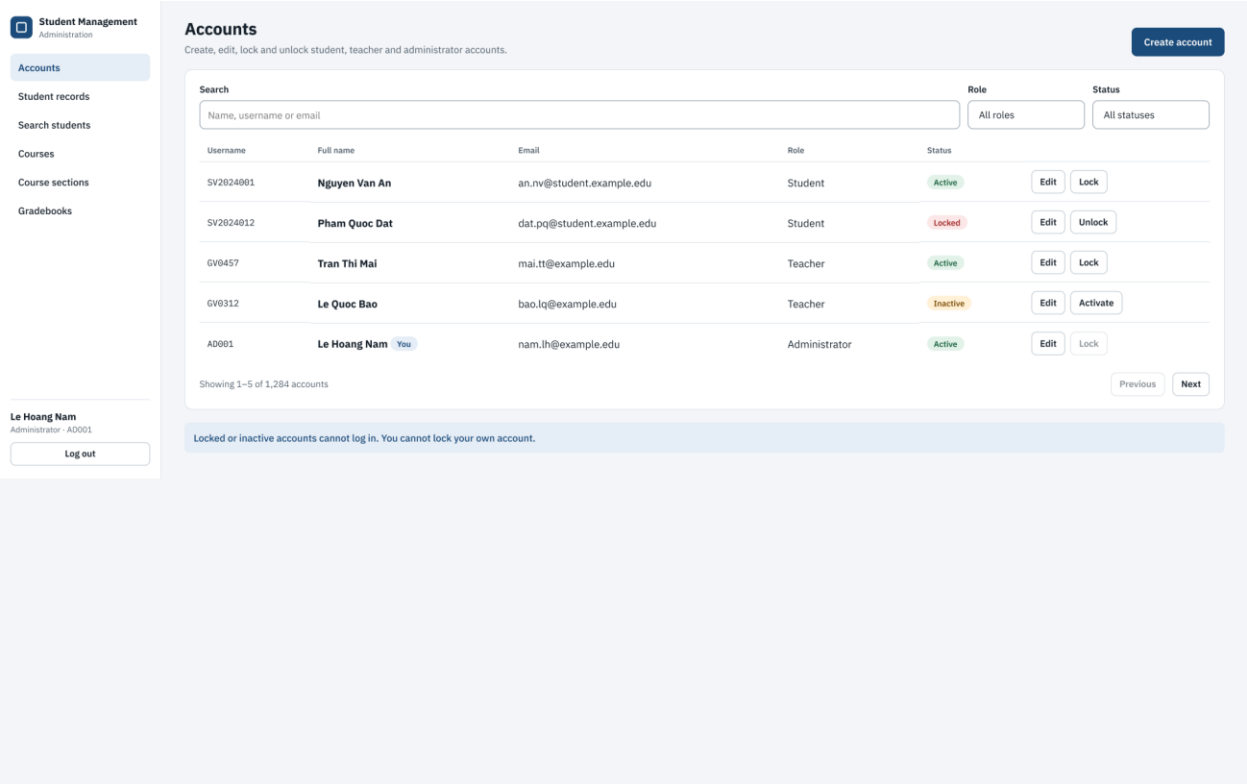


Hình 14:Lock gradebook

After the teacher presses Lock gradebook, a dialog warns that after locking grades can no longer be entered or edited and that they are published to the students. It shows the checks made by the system: grades entered for all students and all grades within 0–10. The teacher confirms with Lock gradebook or closes the dialog with Cancel. If some students still have no grade, the system reports it and asks the teacher to enter all grades before locking.

5.Administrator functions

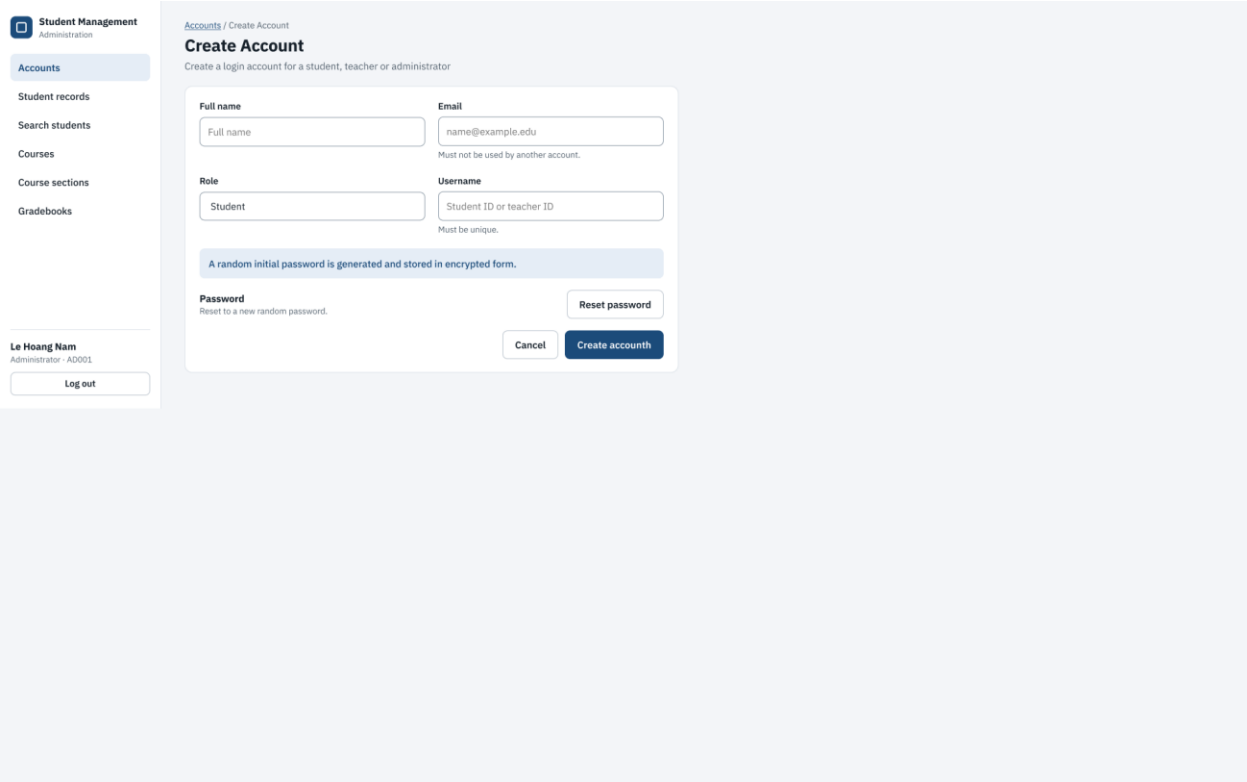
5.1. Accounts



Hình 15:Admin Account Status

This screen lists the accounts of students, teachers and administrators with the username, full name, email, role and status (Active, Locked or Inactive). The Administrator can search by name, username or email and filter by role and status. Each row has an Edit button and a button to change the status: Lock for an active account, Unlock for a locked account and Activate for an inactive account. The Administrator's own account is marked and cannot be locked. Locked or inactive accounts cannot log in. The list is paginated. The Create account button opens the account form.

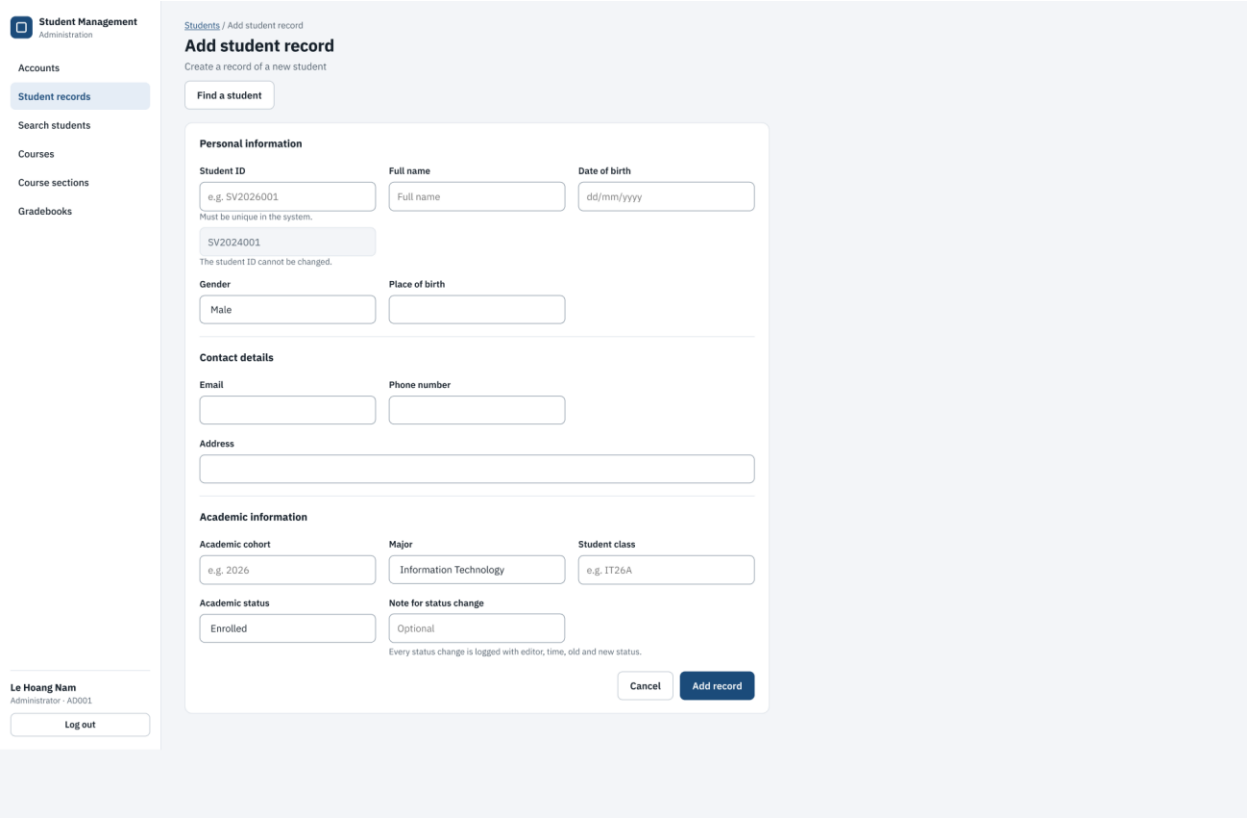
5.2. Create accounts



Hình 16: Admin Create And Edit Account

The Administrator enters the full name, email, role (Student, Teacher or Administrator) and username (student ID or teacher ID) and presses Create account. The email and the username must be unique. The system generates a random initial password and stores it in encrypted form. The same form is used to edit an account, where a Reset password button sets a new random password. If a required field is missing or the email or username already exists, an error message is shown and the Administrator can correct it or cancel.

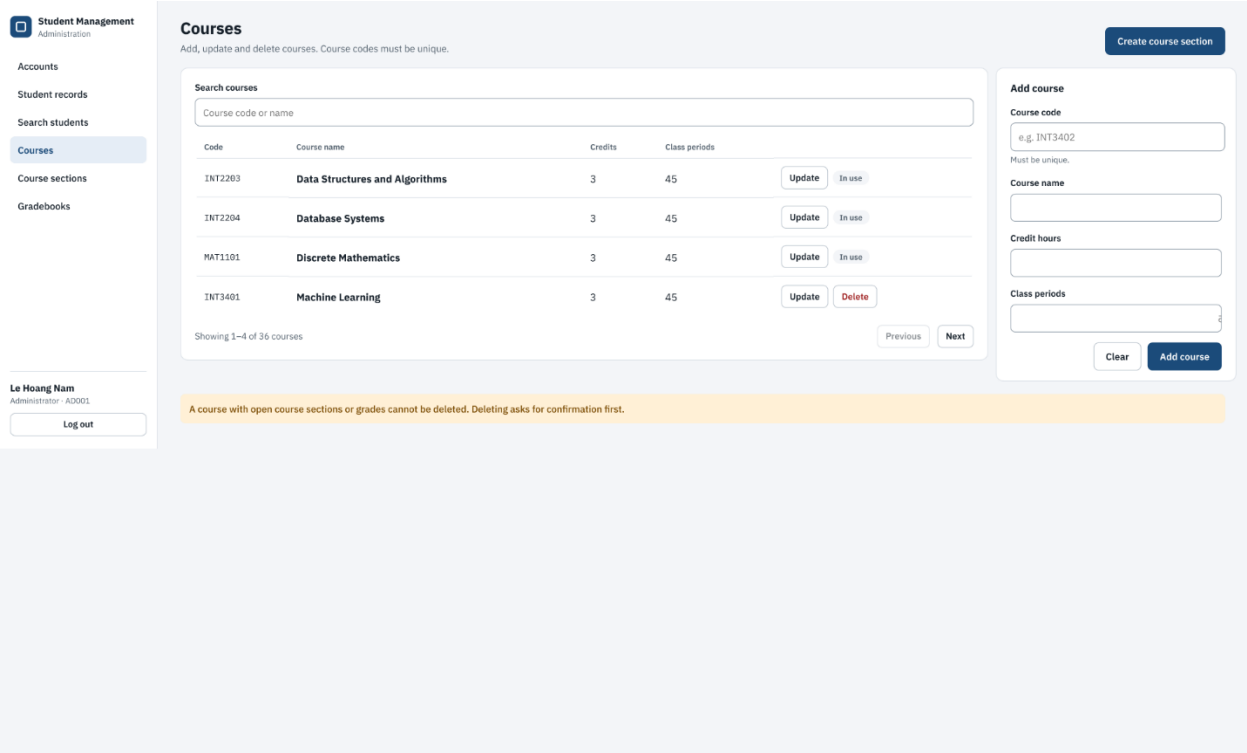
5.3. Student records



Hình 17: Admin Edit Student Record

On the Add student record screen the Administrator enters the personal information (student ID, full name, date of birth, gender and place of birth), the contact details (email, phone number and address) and the academic information (academic cohort, major, student class and academic status), then presses Add record. The student ID must be unique, and missing or invalid data produces an error message. The same form is used to edit a student: the student ID is shown but cannot be changed, and when the academic status is changed the Administrator can enter an optional note; every status change is logged with the editor, time, old status and new status. The Find a student button is used to look for the student to edit.

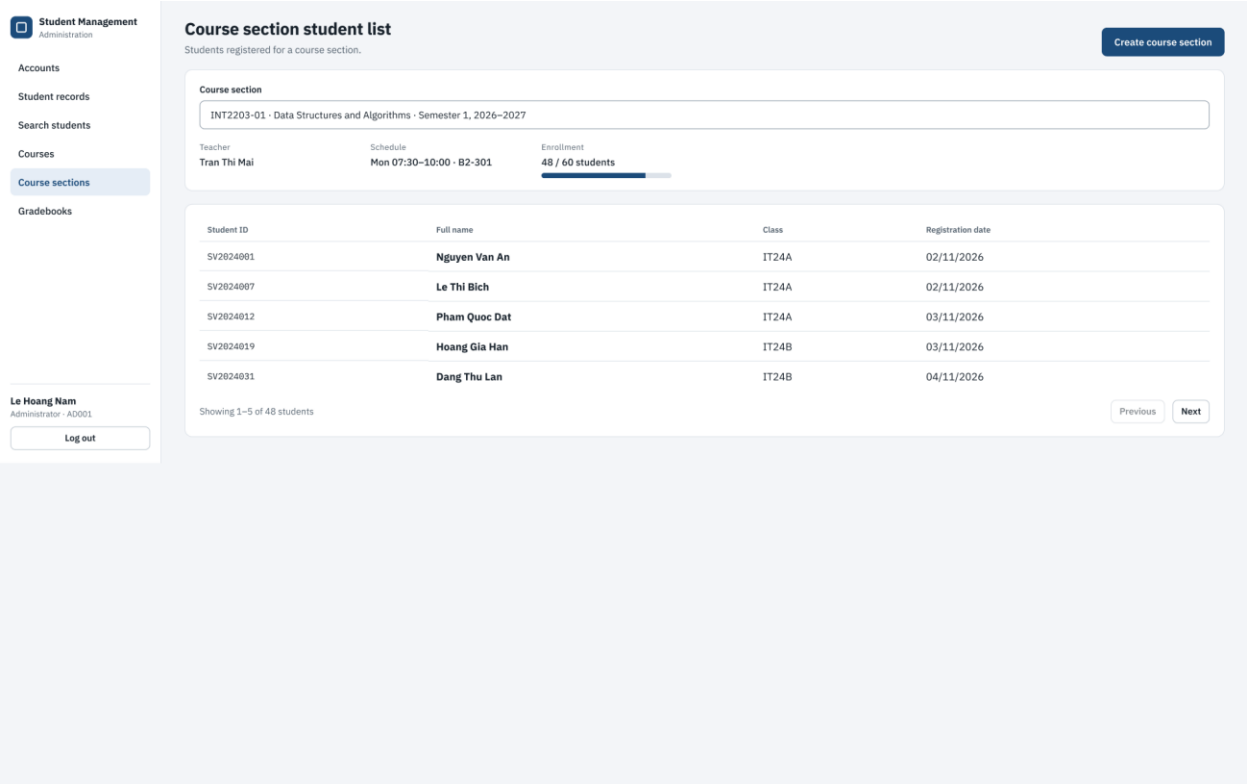
5.4. Courses



Hình 18: Admin Manages Course

This screen lists the courses with the code, name, credits and class periods, and can be searched by course code or name. Each course has an Update button; a course that is not used also has a Delete button, which asks for confirmation, while a course in use shows the label In use and cannot be deleted. Courses that have open course sections or grades cannot be deleted. New courses are added with the Add course form on the right (course code, course name, credit hours and class periods) and the Clear button empties the form. The course code must be unique. The list is paginated, and a Create course section button leads to the section creation form.

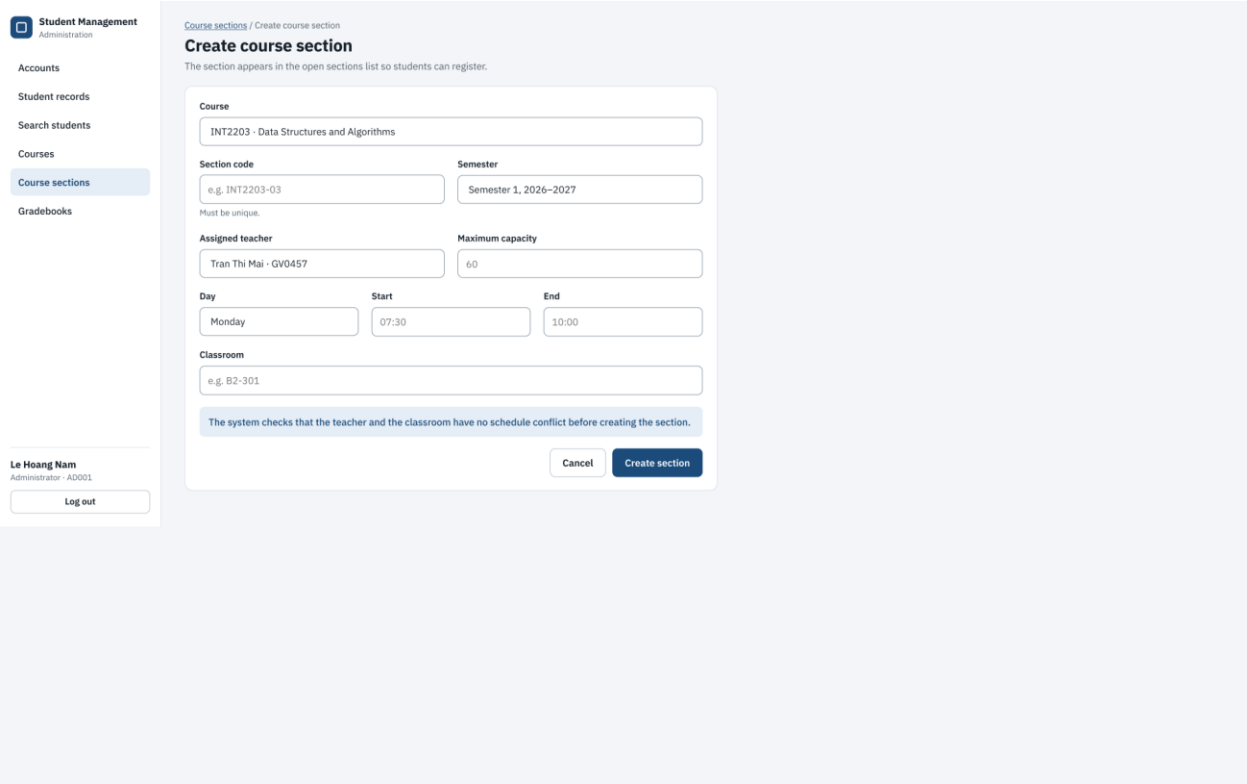
5.5. Course section student list



Hình 19:Admin Course Student List

The Administrator chooses a course section and the screen shows its teacher, schedule and room, and the enrollment against the maximum capacity. The list shows the registered students with their student ID, full name, class and registration date, with pagination. A Create course section button opens the section creation form. If no student has registered, a message is shown.

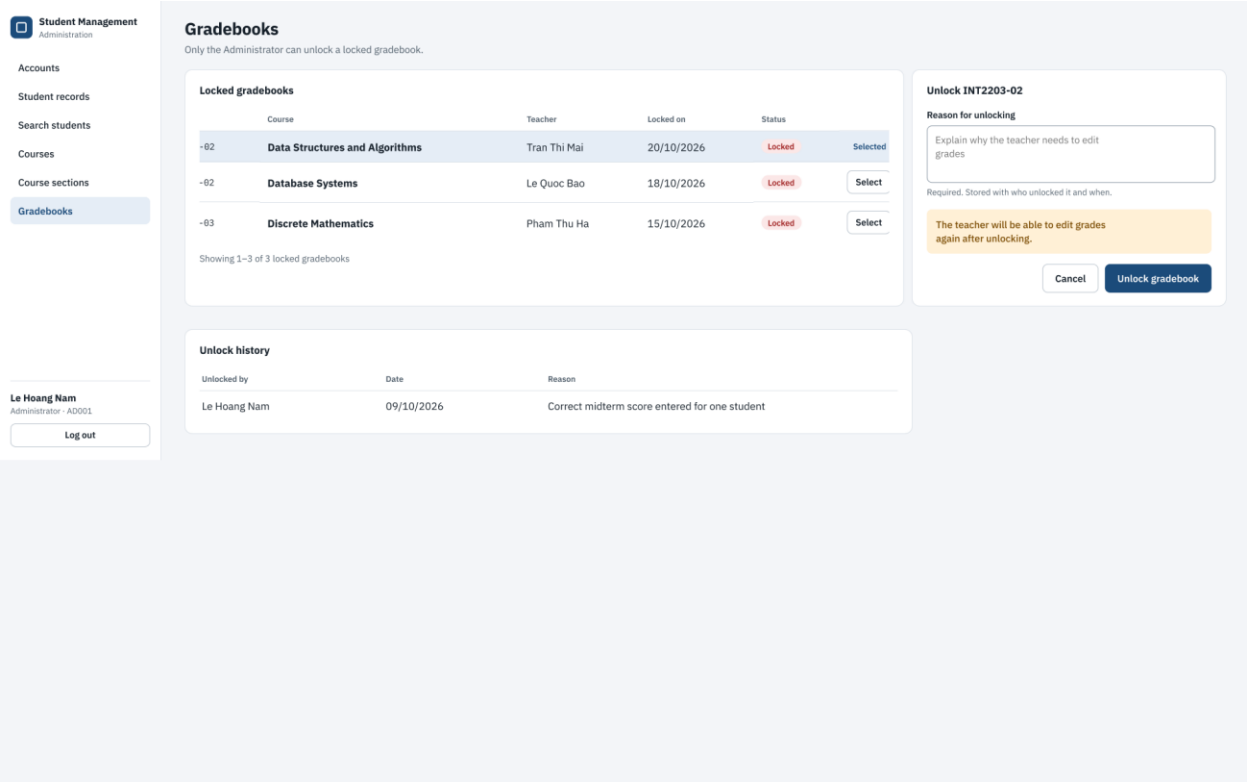
5.6. Create course section



Hình 20:Admin Create Course Section

The Administrator chooses the course and enters the section code, semester, assigned teacher, maximum capacity, day, start and end time and classroom, then presses Create section. The section code must be unique. The system checks that the teacher and the classroom have no schedule conflict before creating the section; if there is a conflict or the code already exists, an error message is shown and the Administrator can change the data or cancel. The new section then appears in the open sections list so that students can register.

5.7. Gradebooks



Hình 21: Admin Gradebook

This screen lists the locked gradebooks with the section, course, teacher and the date they were locked. The Administrator selects a gradebook, enters the required reason for unlocking and presses Unlock gradebook, or presses Cancel to clear the selection. The system warns that the teacher will be able to edit grades again after unlocking, asks for a reason if none is entered, and records who unlocked it, when and why. An unlock history lists who unlocked a gradebook, the date and the reason.